

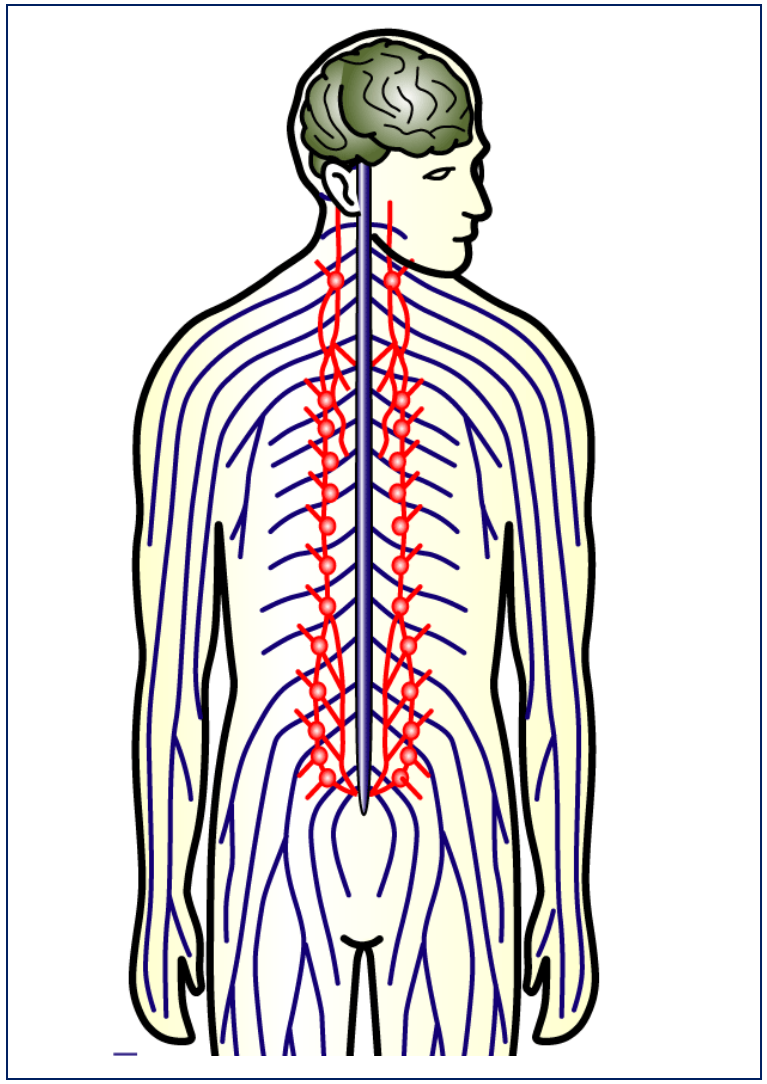
Neuro-mechanics of human movement and modeling

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Today’s plan

- Motor System
- Neuromusculoskeletal Model
- Group exercise

The motor system



Bend your arm!

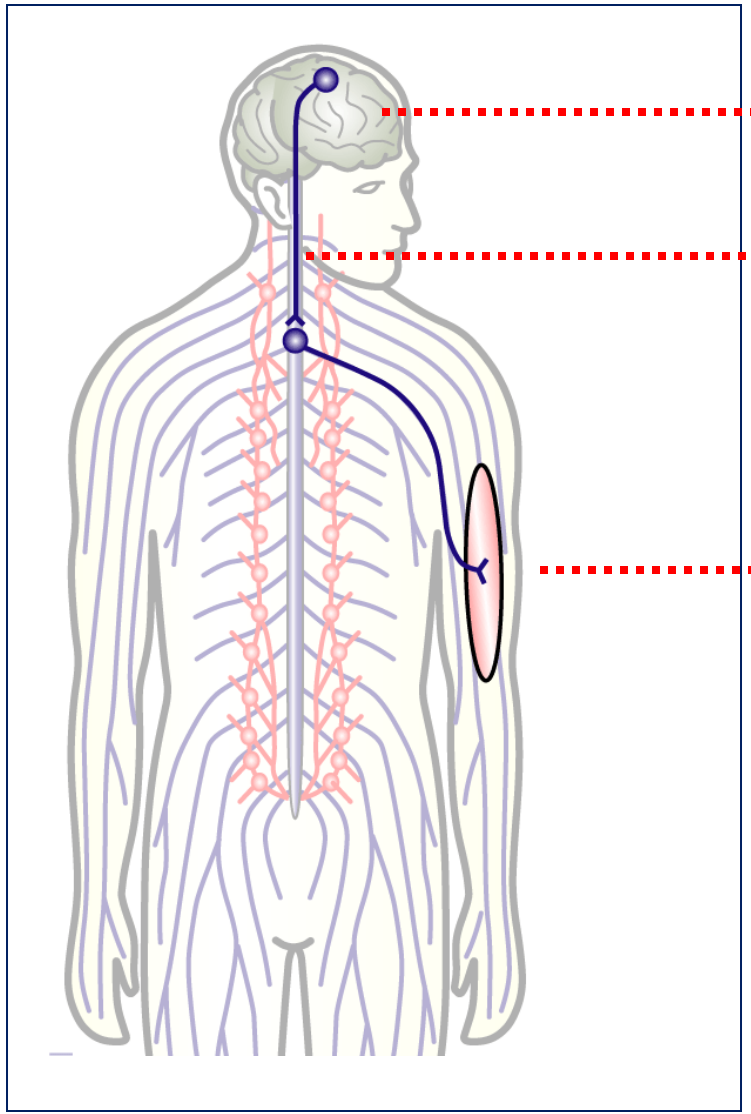
What happens from making the decision to the execution of the movement?

Where does it happen?

How long is the total response time?

Discussion with your neighbors for 30s

The motor system



Brain think / decide

Spinal cord transmit / pass decision on

Muscles respond / contract

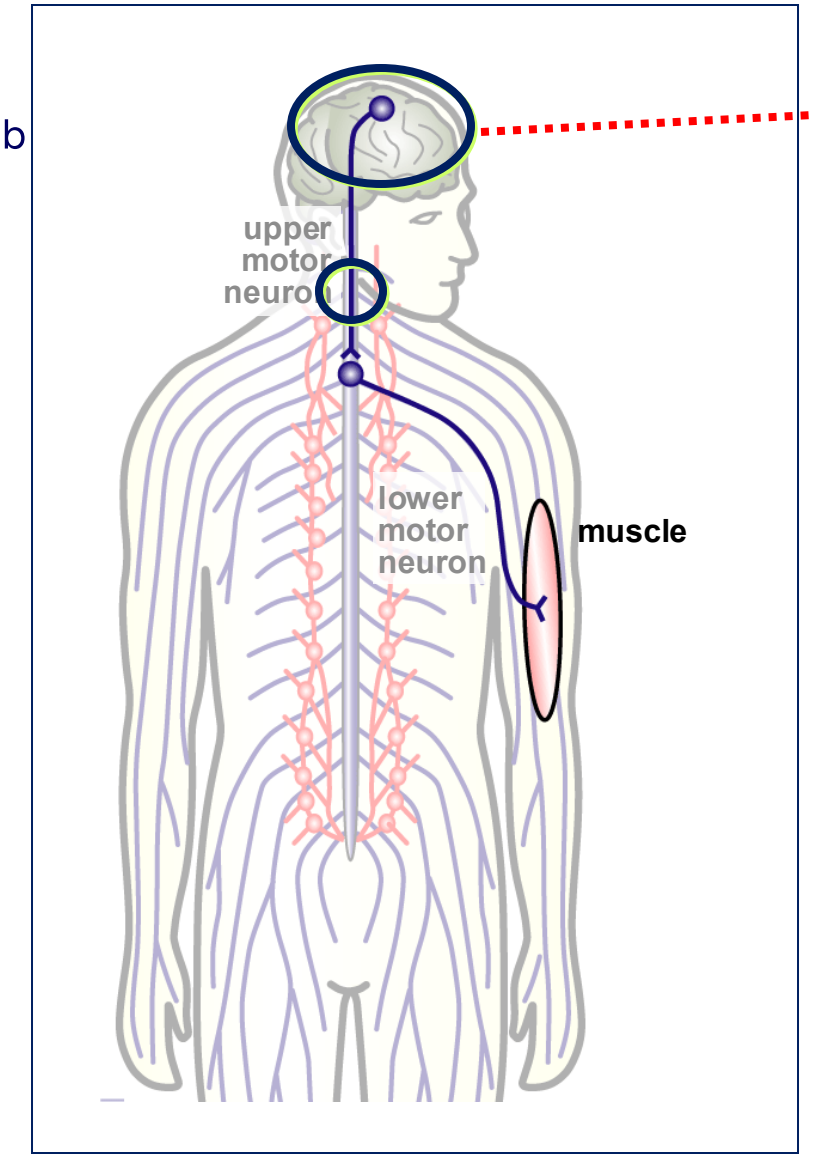
How does your nervous system distinguish between bending & straightening your arm?

Total response time:
~ 150-250ms

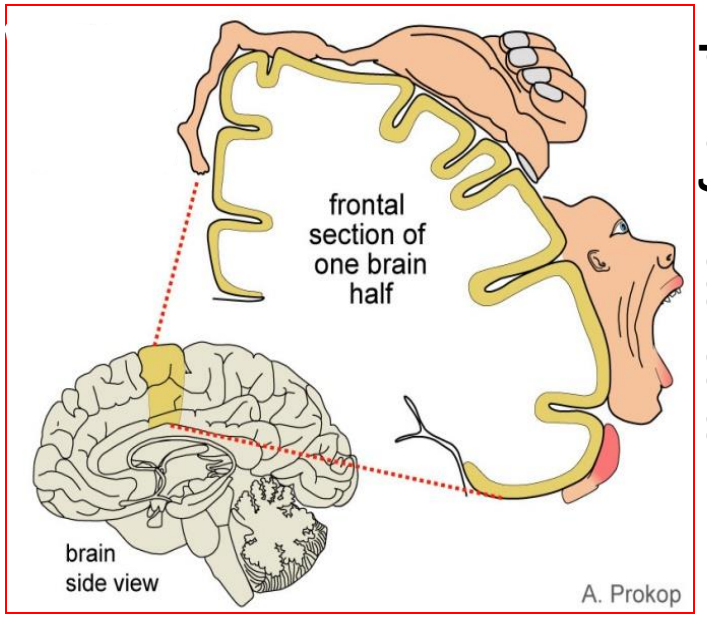
- Reaction time: ~ 120ms
- Neural Conduction Time: ~10-40 ms
- Electromechanical Delay (EMG to Force): ~30-100 ms

The motor system

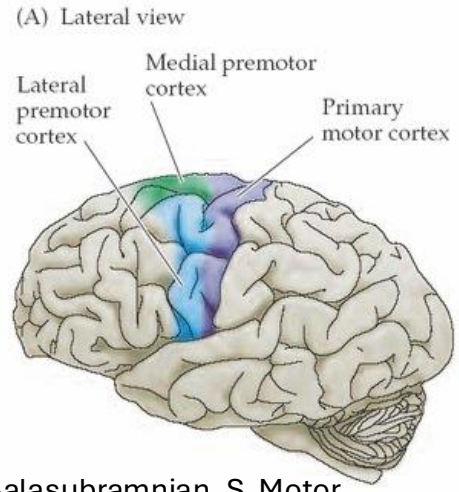
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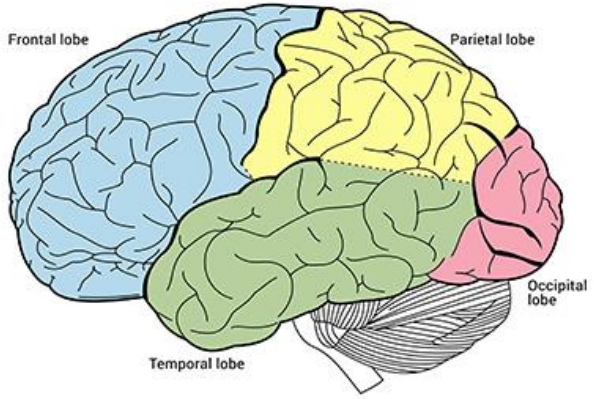
Homunculus map/little man

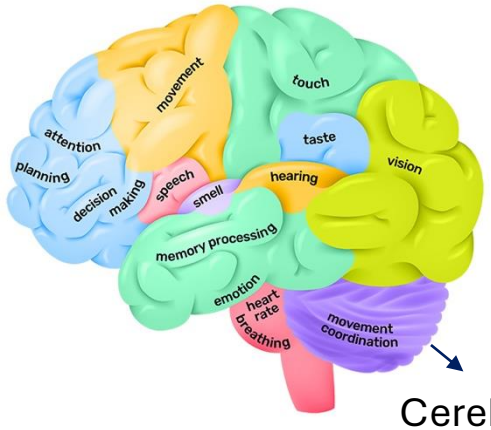


primary motor cortex



Balasubramnian, S. Motor impairment following stroke, 2015

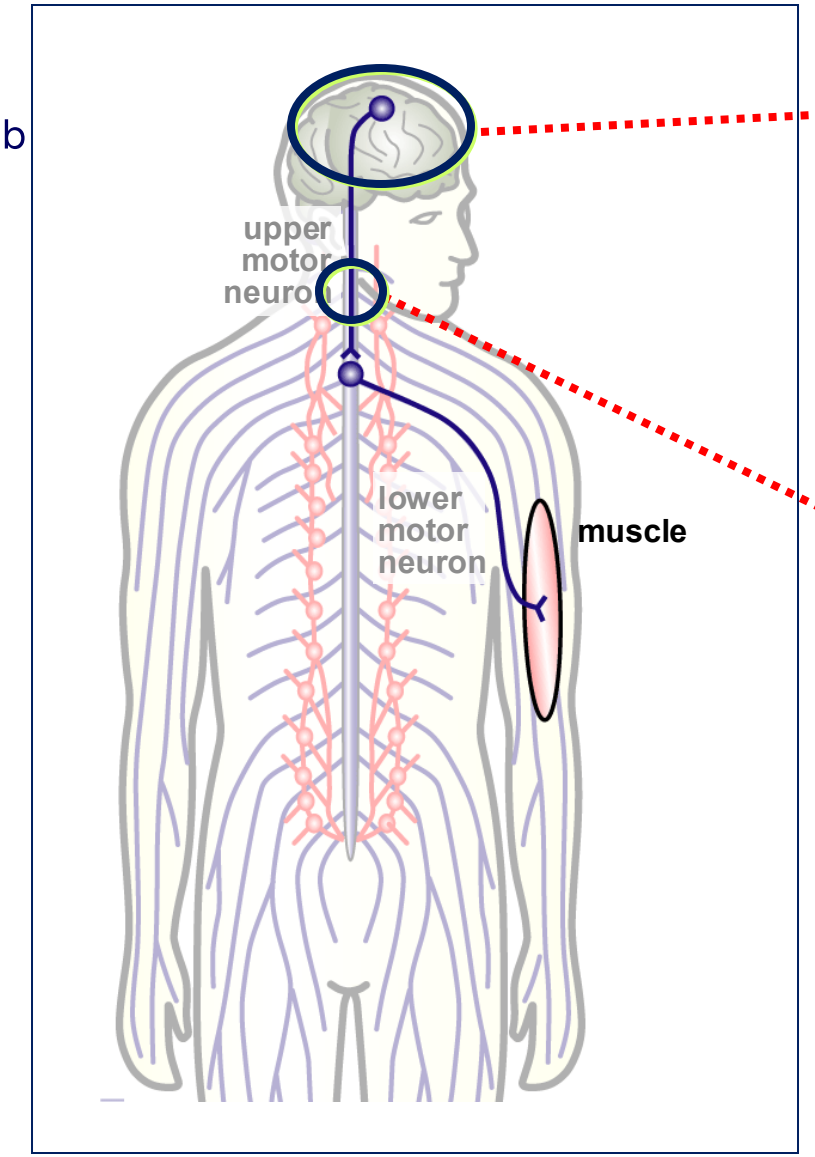


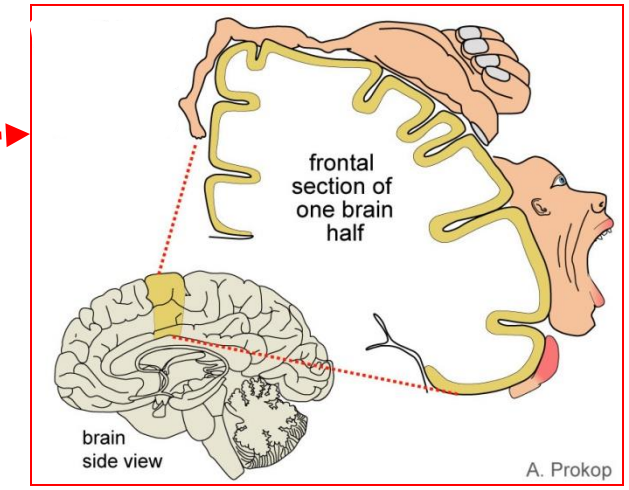


Cerebellum

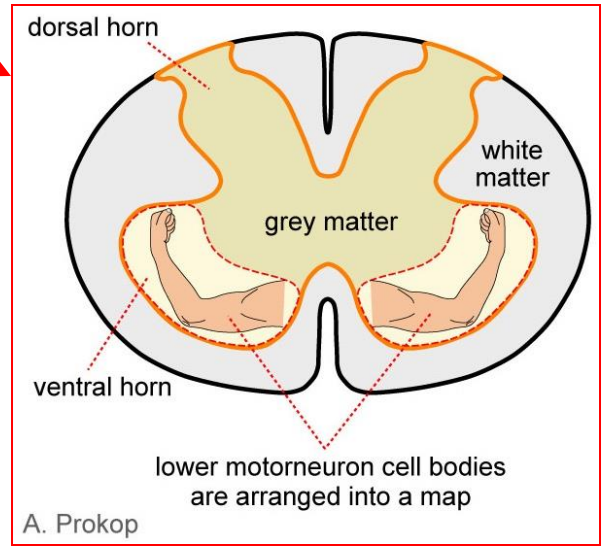
wikimedia

The motor system

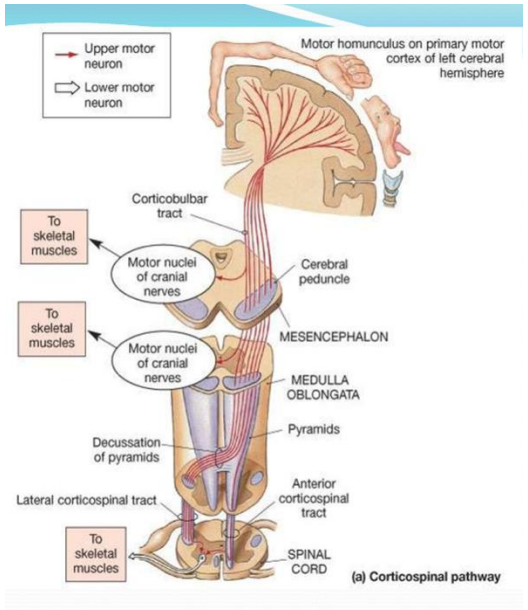




primary motor cortex



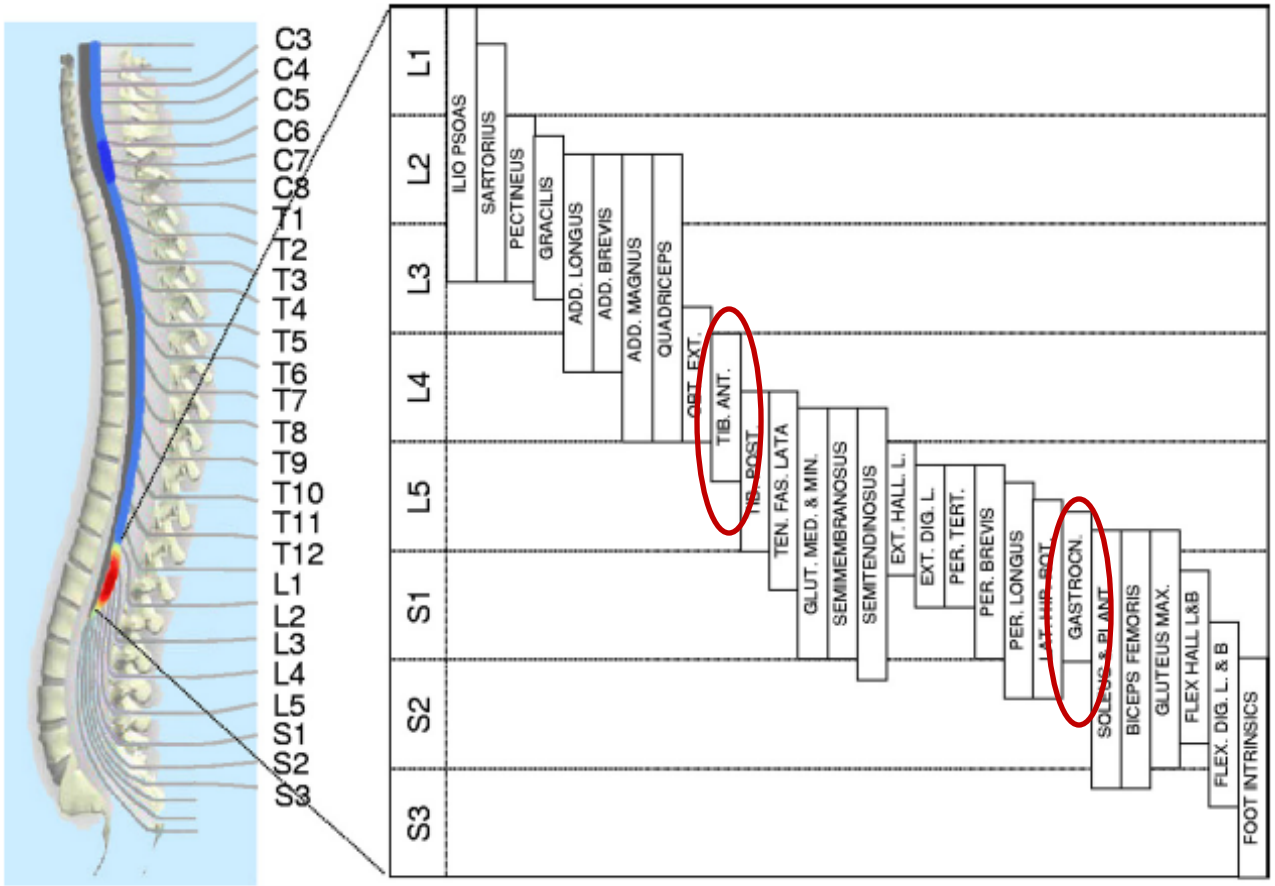
spinal cord cross-section



There are maps in primary motor cortex and spinal cord

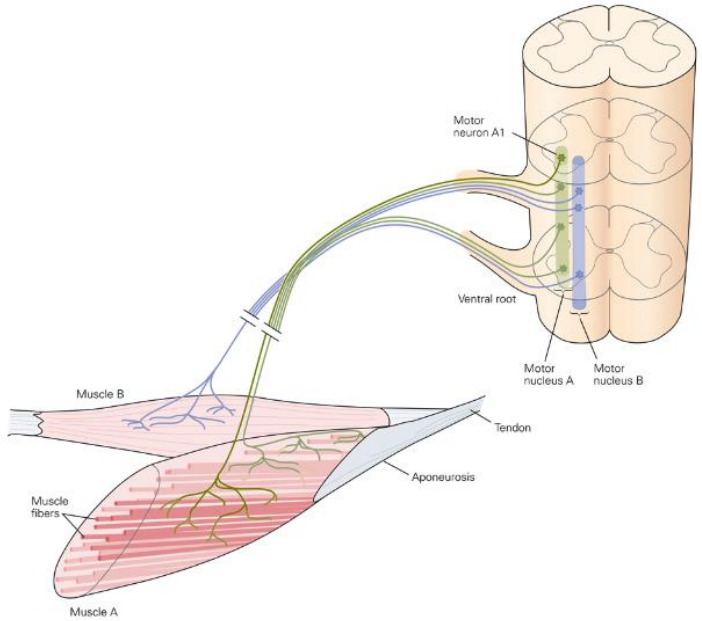
The motor system

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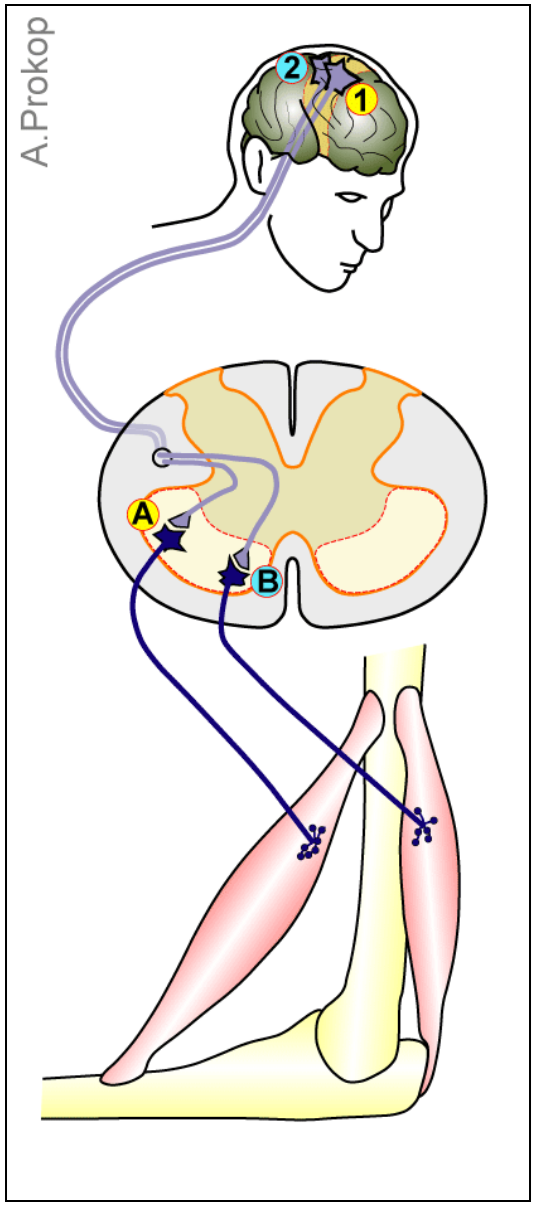


Germana Cappellini et al., J Neurophysiol 104,2010

Cervical -> respiration, head, neck and arm
Thoracic -> finger, chest, back, abdominal
Lumbar and sacral -> locomotion, bowel, reproductive function

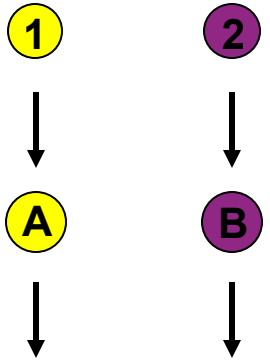


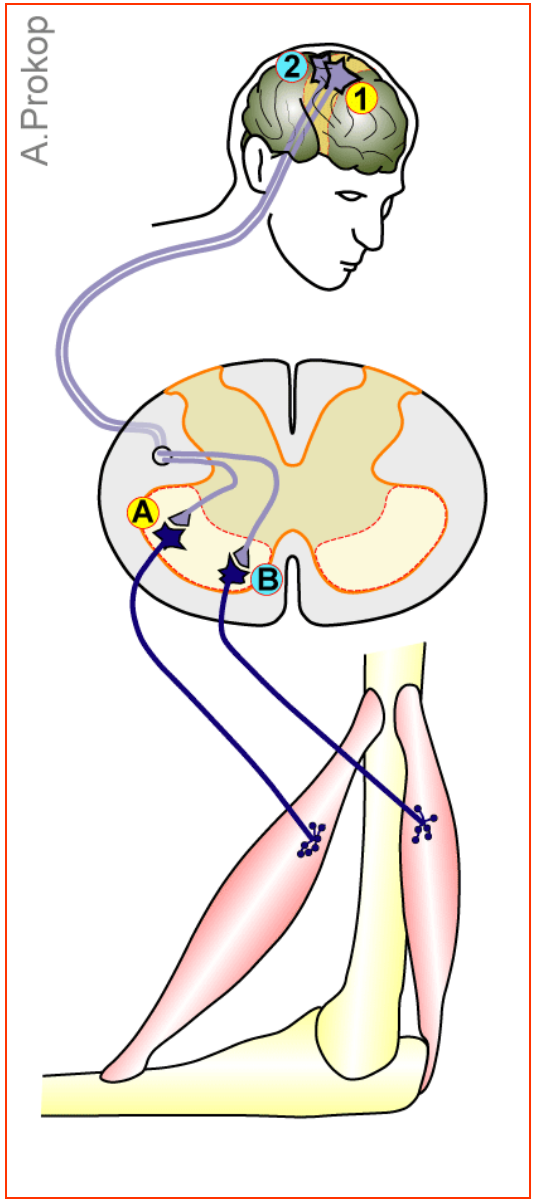
Jun-An Chen, Neuroscience: Exploring the brain, IMB, Academia Sinca



What happens when stimulating upper motoneuron

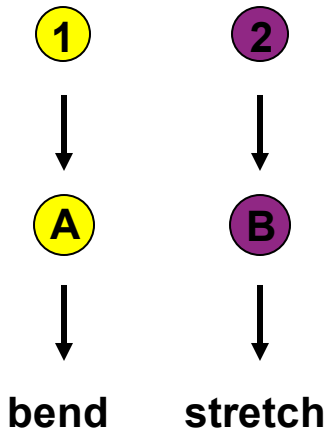
1 *versus* **2** ?





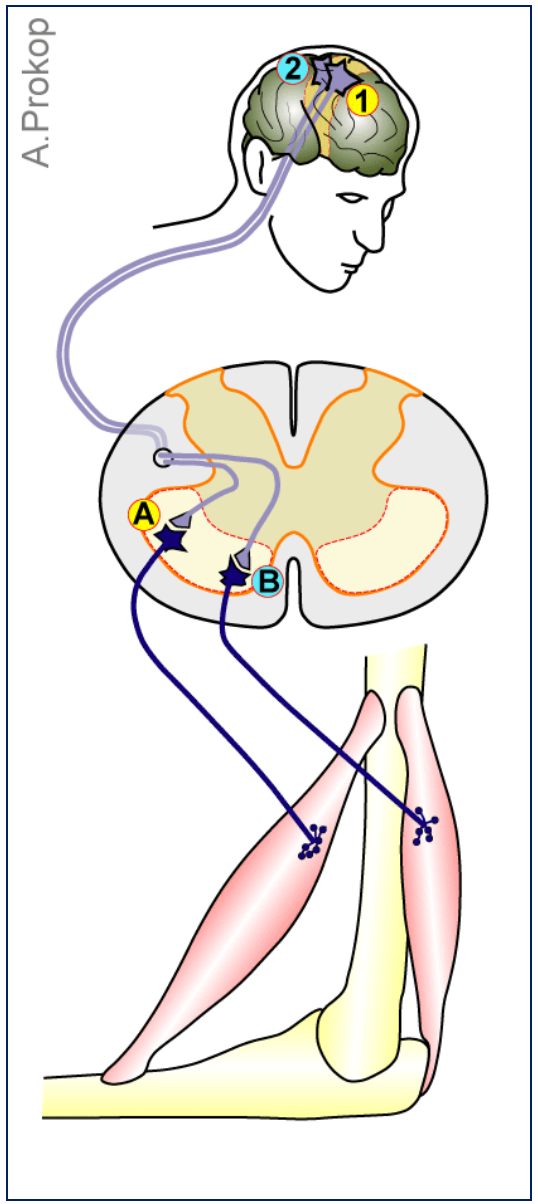
What happens when stimulating upper motoneuron

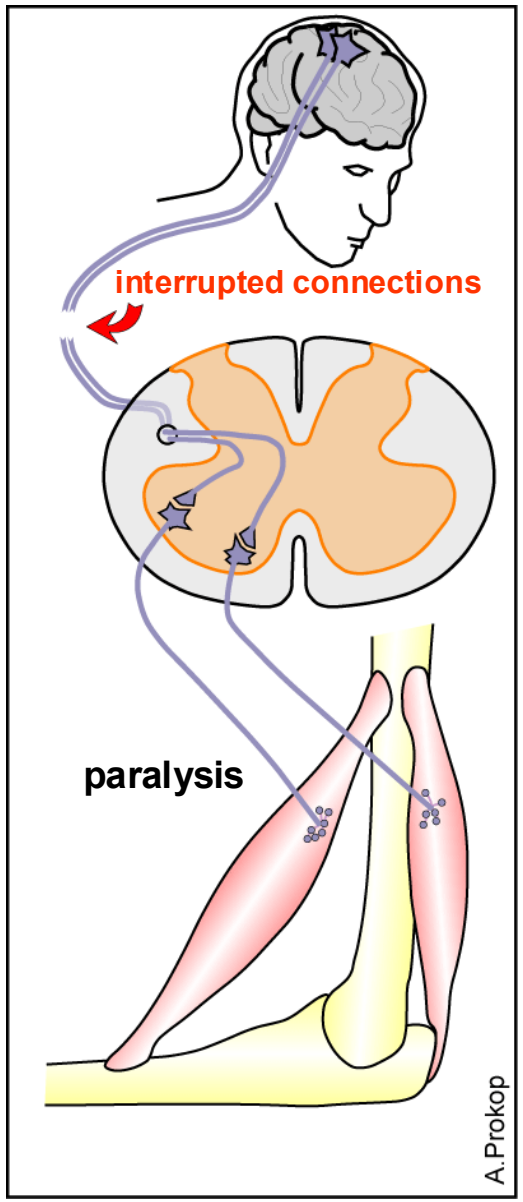
1 *versus* **2** ?

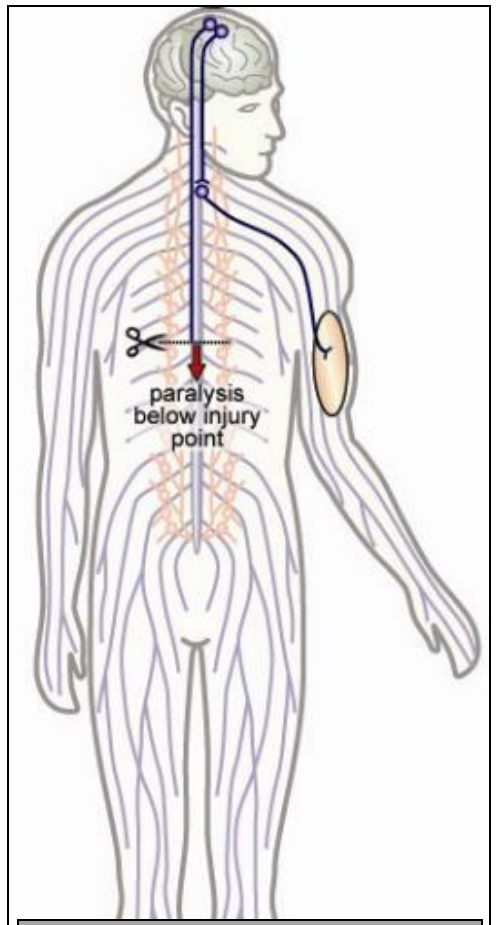


What happens in persons with neurological disorders?

The motor system





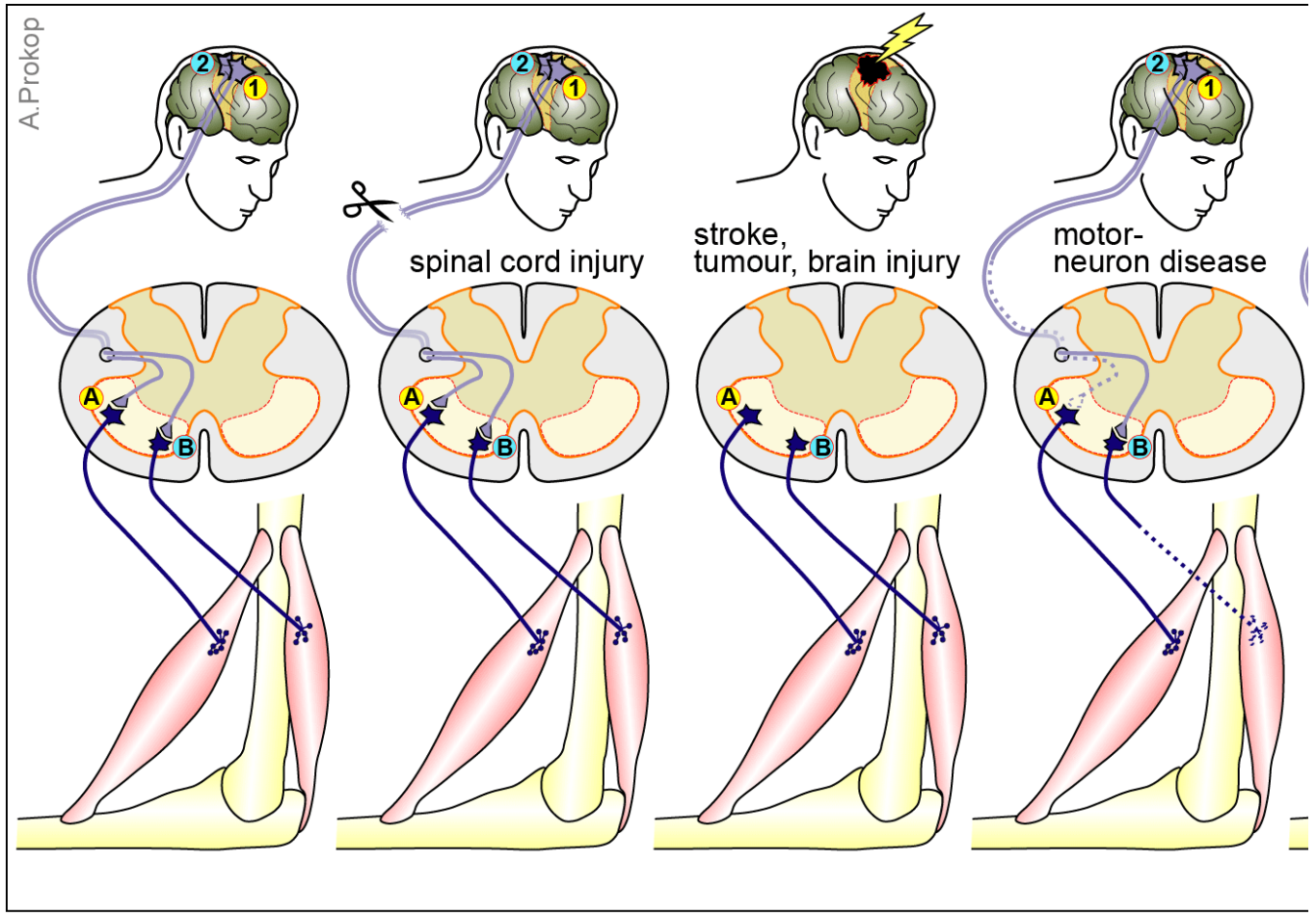


Paralysis below the injury site, showing the importance of the nerve connections.

- Global incidence: 40-80 cases per million population per year
- Total new cases globally: ~250,000-500,000 per year

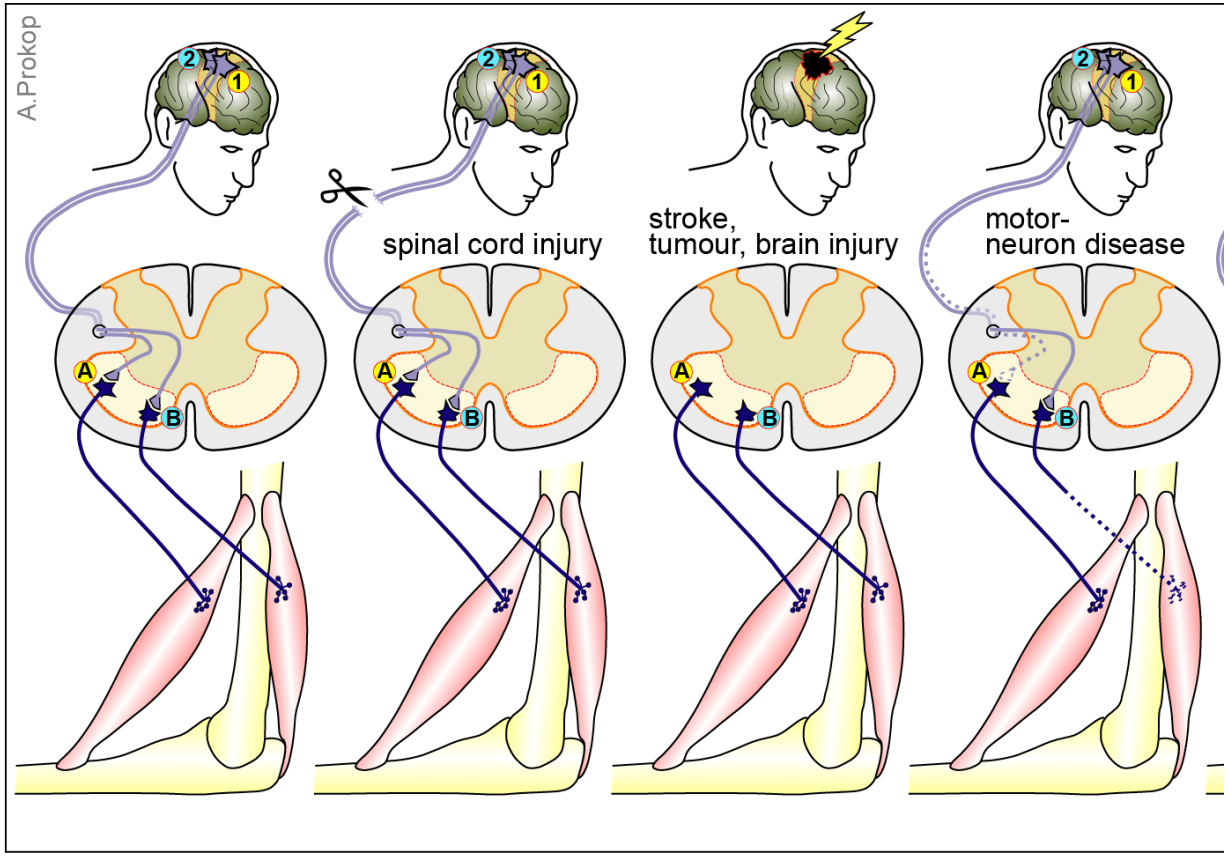
Kumar, R., et al. (2018). *World Neurosurgery*

The motor system



The motor system

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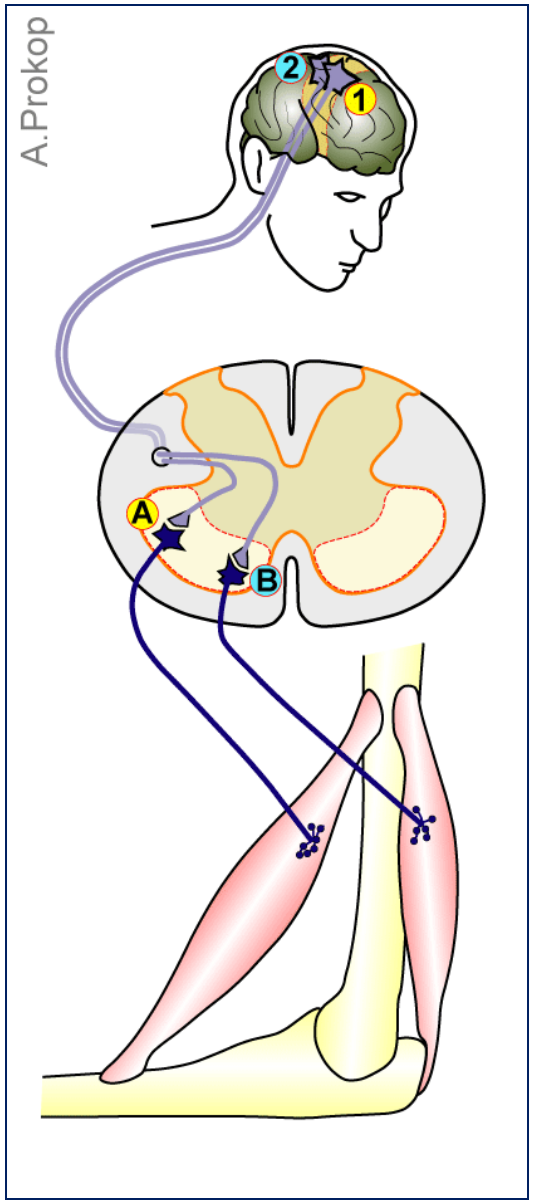


Conditions	SCI	Stroke	Motor Neuron Disease (ALS)
Site of Lesion	Spinal cord	Brain	Progressive death of upper and lower MNs
Neural pathway	Upper MN	Upper MN	Upper and lower MN
Voluntary contraction	Lost (partial) below injury level	Impaired contralateral side	Progressive loss throught the body
Sensation	Below lesion	Maybe impaired on the affected side	normal

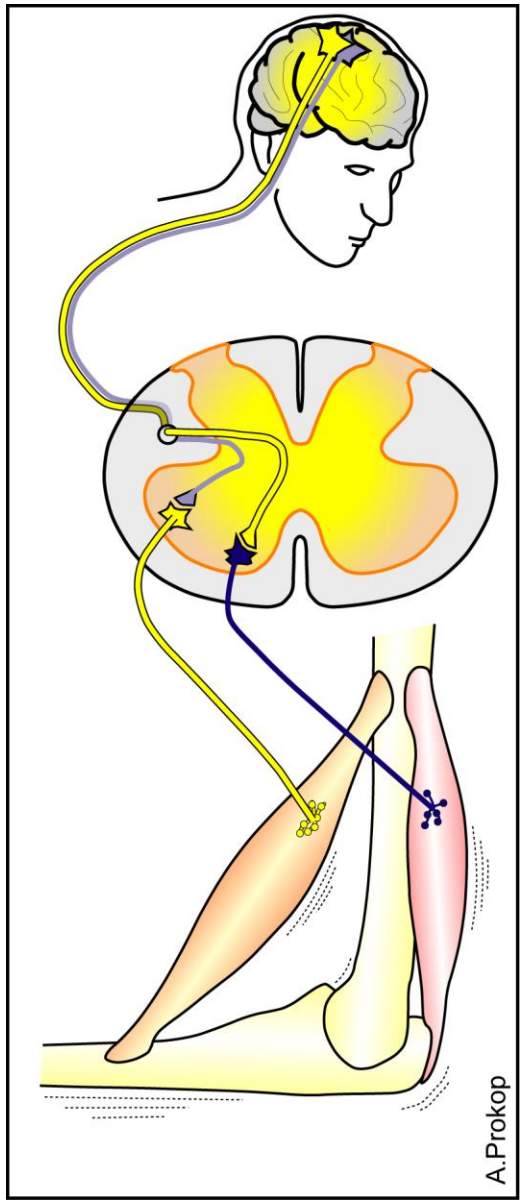
The motor system

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What would happen if all neurons fired at the same time?

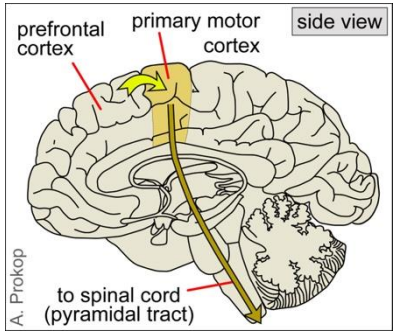


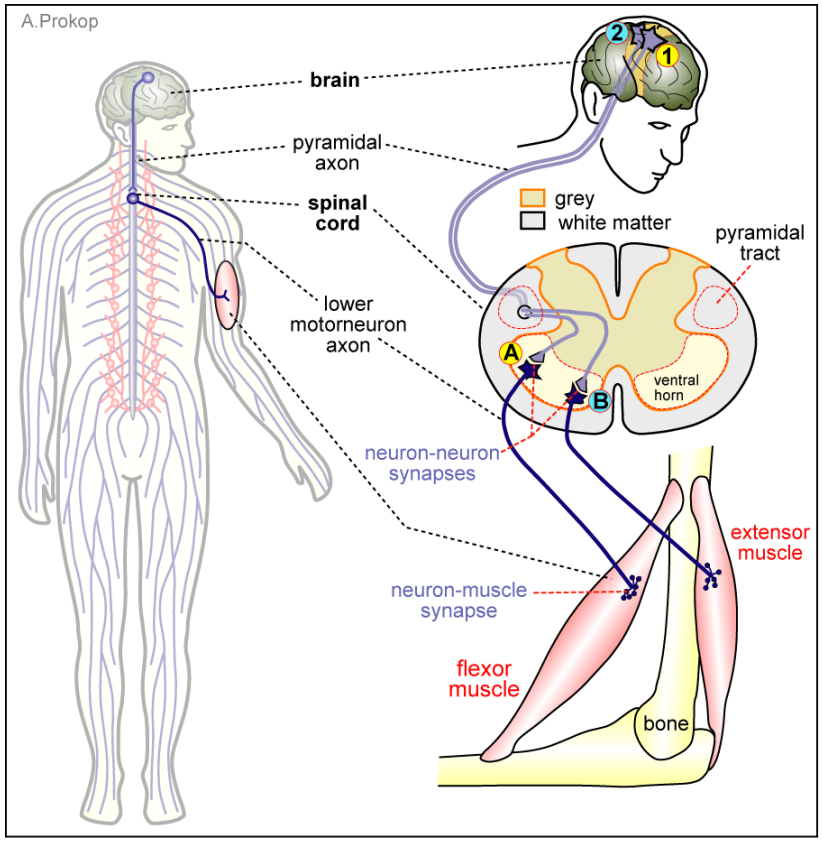
epileptic seizure



Bend your arm

1. The “decision making” occurs in the nerve networks of your brain, e.g., the prefrontal cortex
2. The pyramidal axons (**upper motoneurons**) send **nerve impulses** (the message) all the way down from the brain to the spinal cord, and connect to **lower motoneurons** (A and B)
3. Lower motoneurons project their axons through peripheral nerves to form synapses (neuron-muscle) with elbow extensors and flexors in the arm
4. Muscle contraction -> Force (**number of motor units** and **rate of discharge action potential**) -> Movement

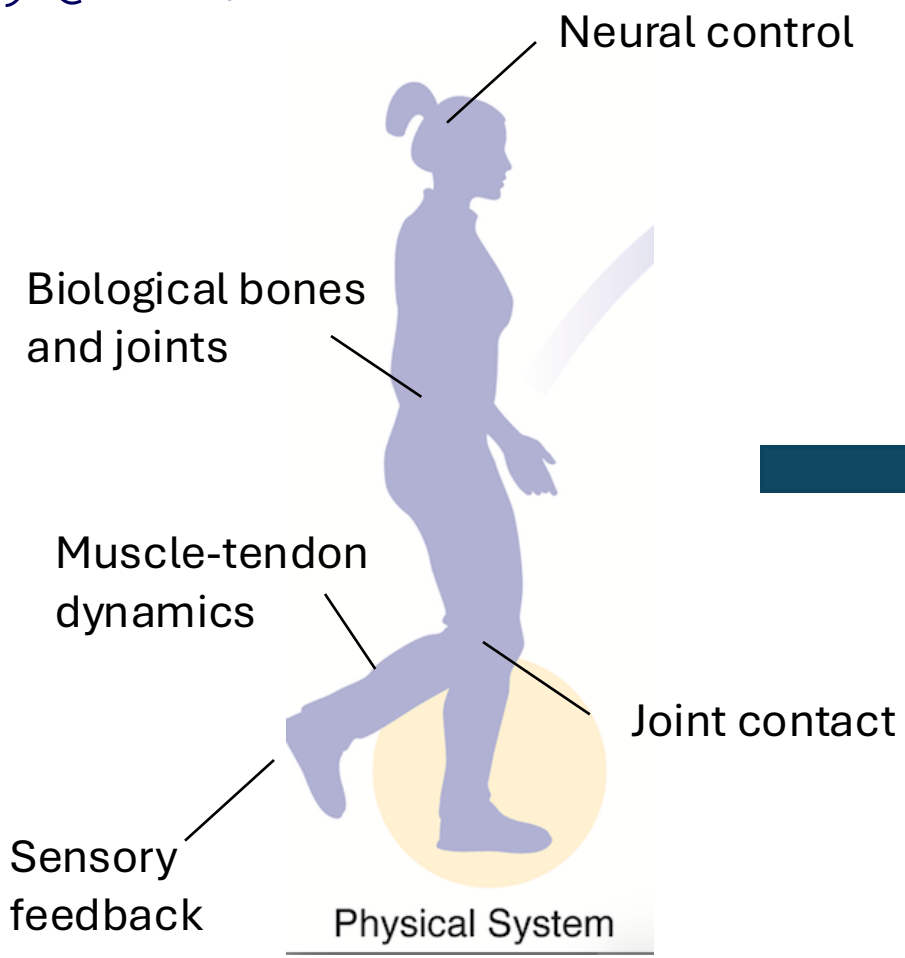




Manchester Fly Facility (2015). droso4schools: Online resources for school lessons, https://droso4schools.wordpress.com/

Neuro-musculoskeletal model

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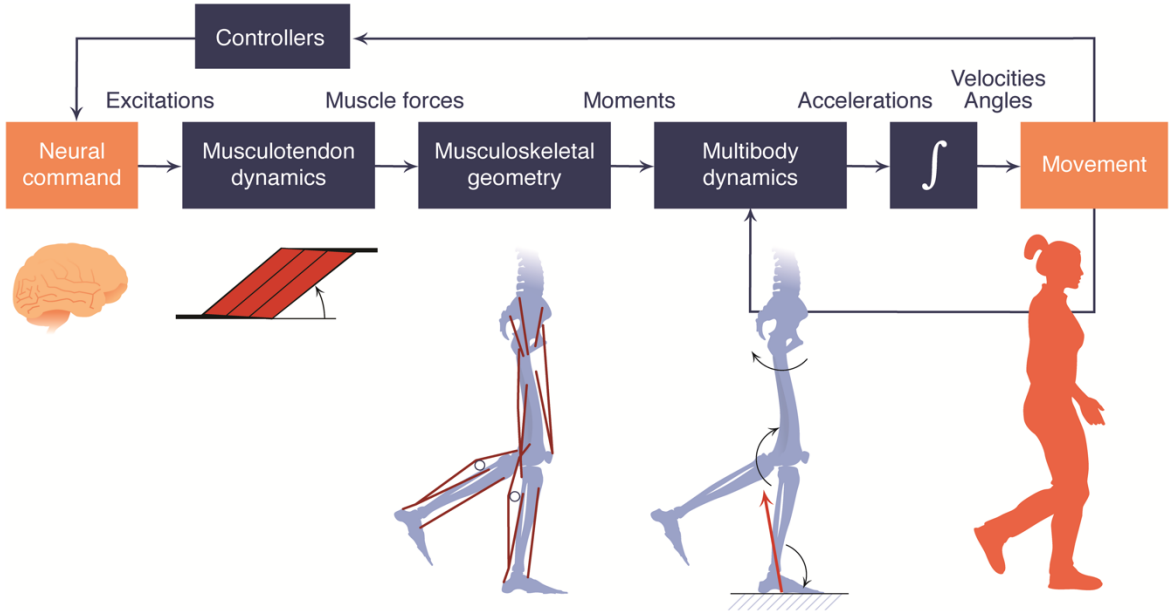


Neurophysiological level

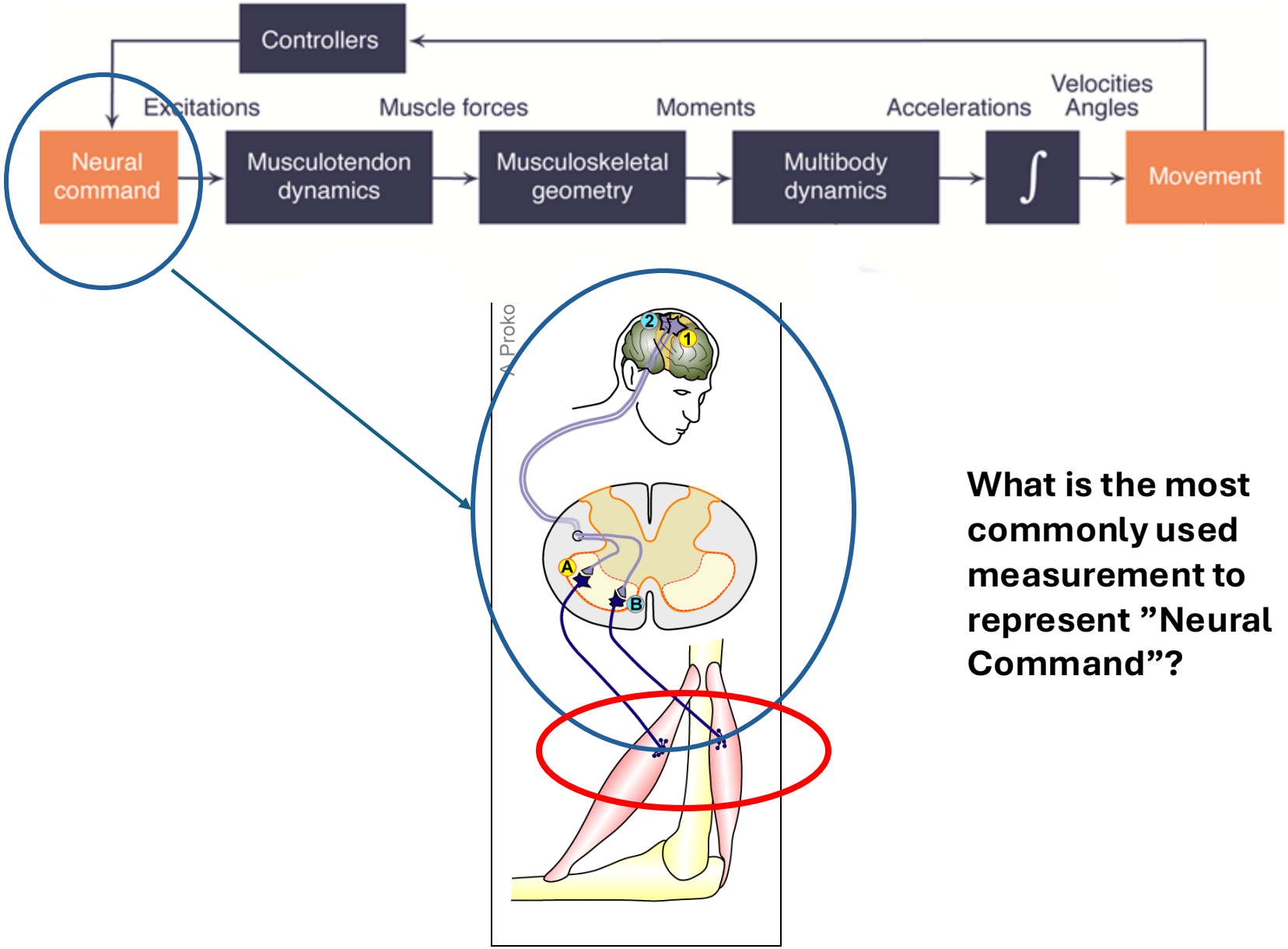


Musculoskeletal level





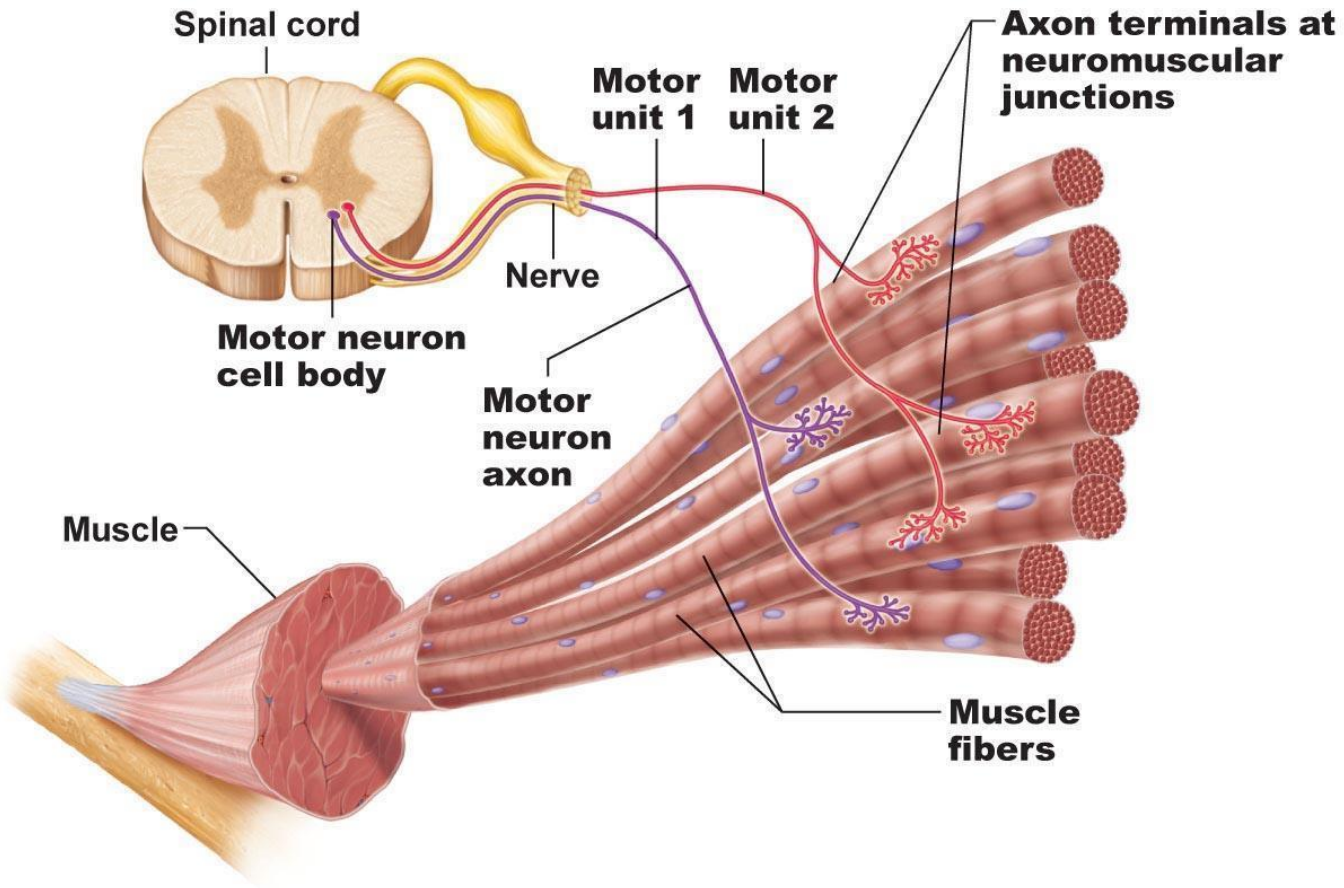
Seth A. et al., Plos Computational Biology, 2018



What is the most commonly used measurement to represent ”Neural Command”?

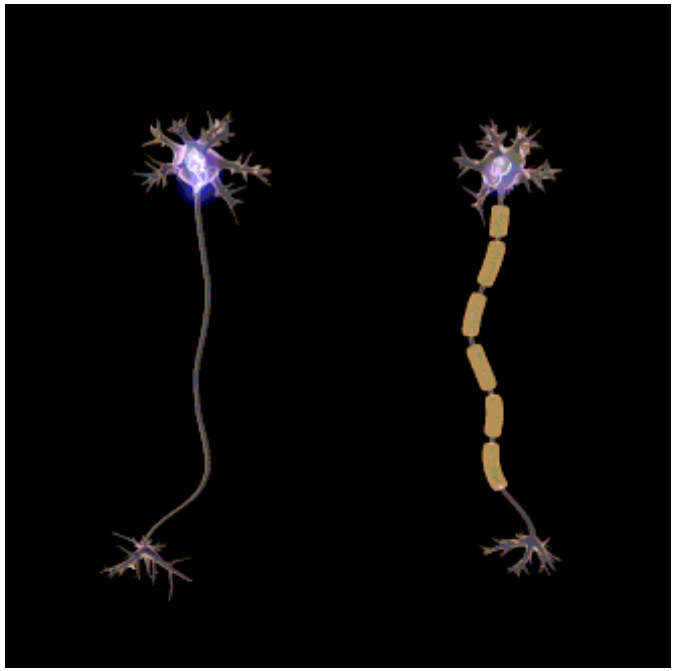
Neural Command - Motor Unit (MU)

- **MU**: A basic **functional** unit of the neuromuscular system, comprises a motor neuron and muscle fibers innervated by the axon
- **Motor neuron pool (MNP)**: the population of motor neurons that innervate a single muscle



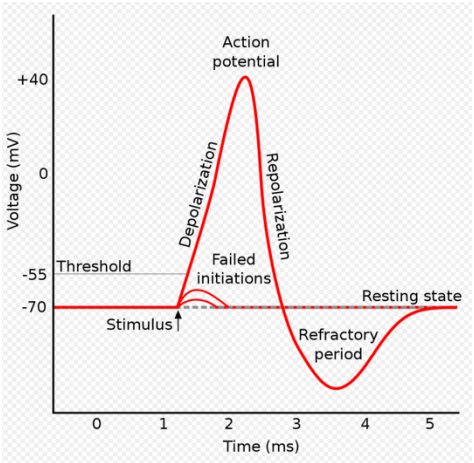
courtesy to Enrico Merlo, OT Bioelettronica for the illustration

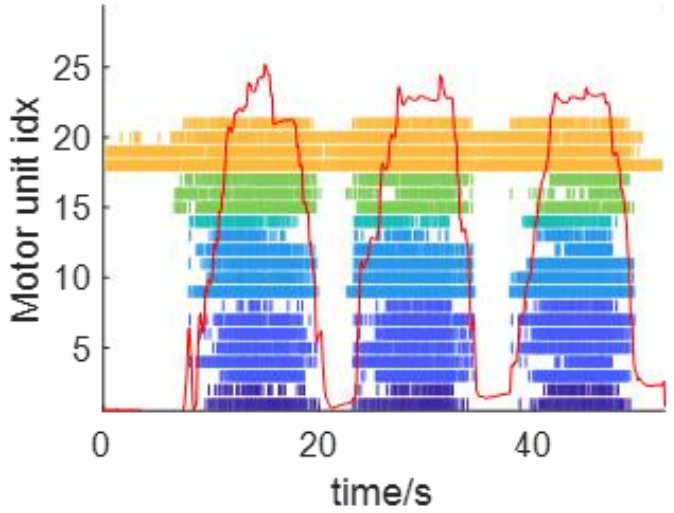
Action Potential (AP)



AP propagation from node to node (Wikipedia)

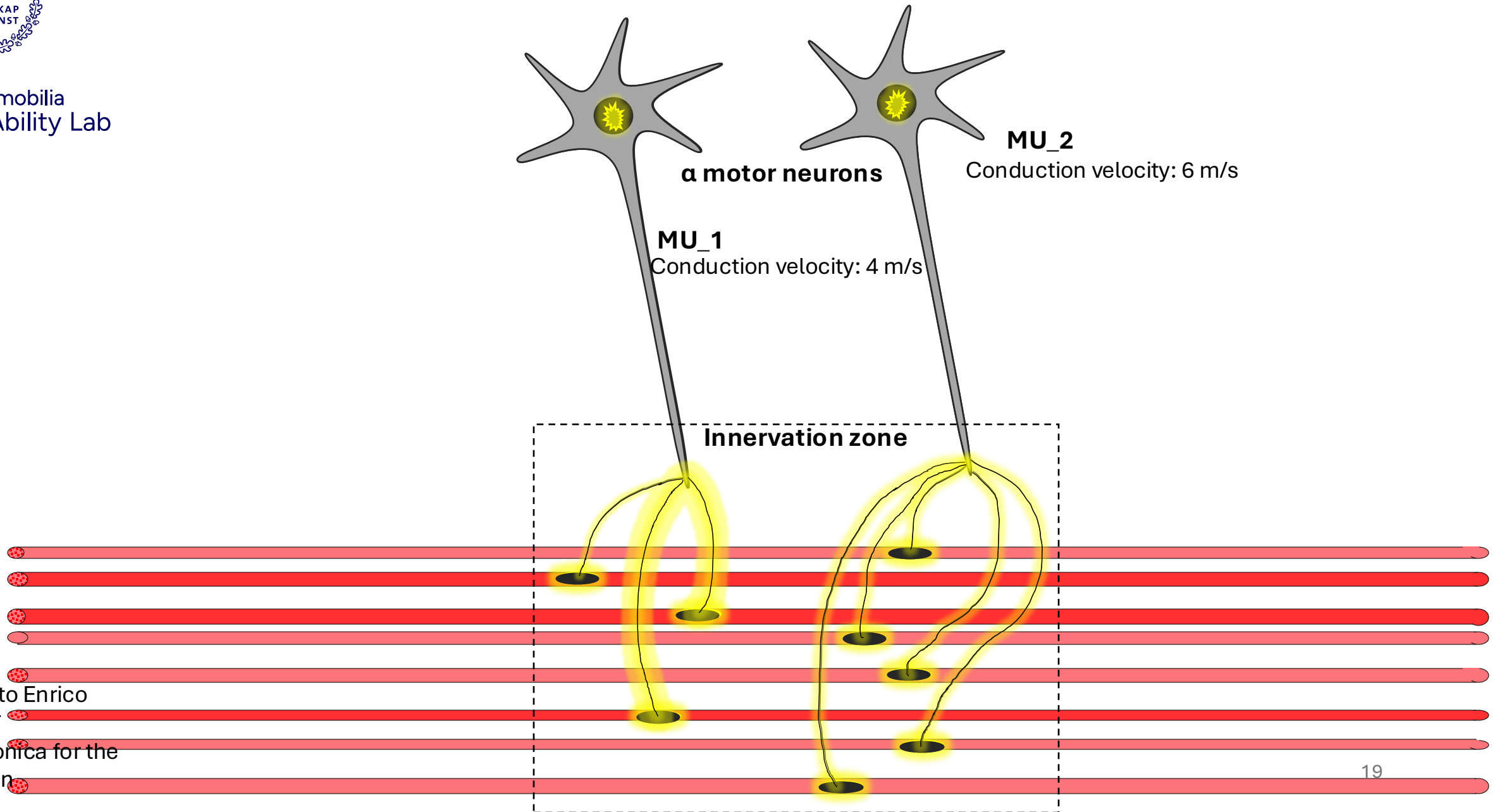
- A rapid and subsequent fall in voltage
- Happens when a neuron sends information down an axon
- Caused when different ions across the neuron membrane
- All APs are the same size -> firing is either on or off







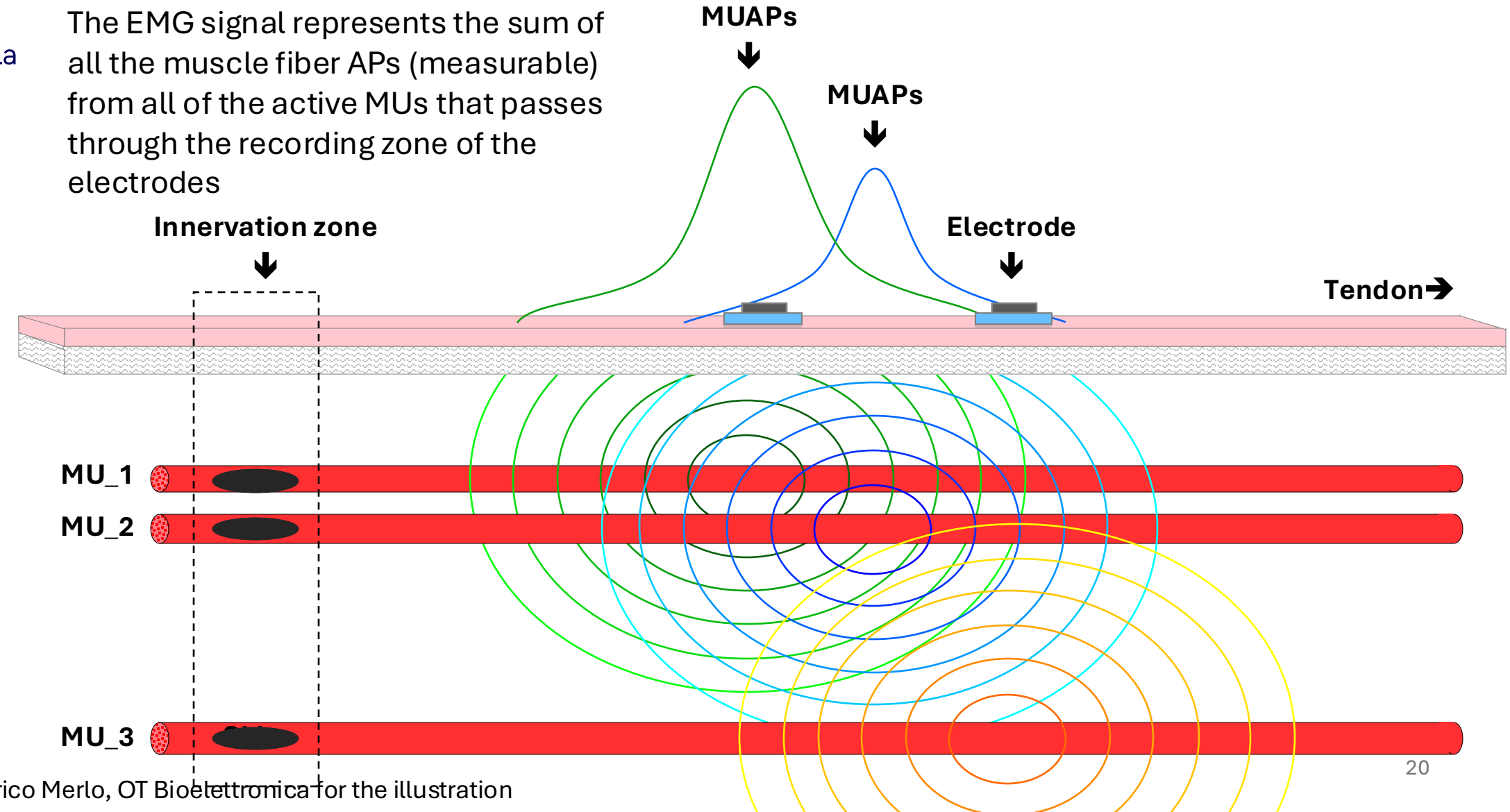
Motor Units Action Potentials



courtesy to Enrico Merlo, OT Bioelettronica for the illustration

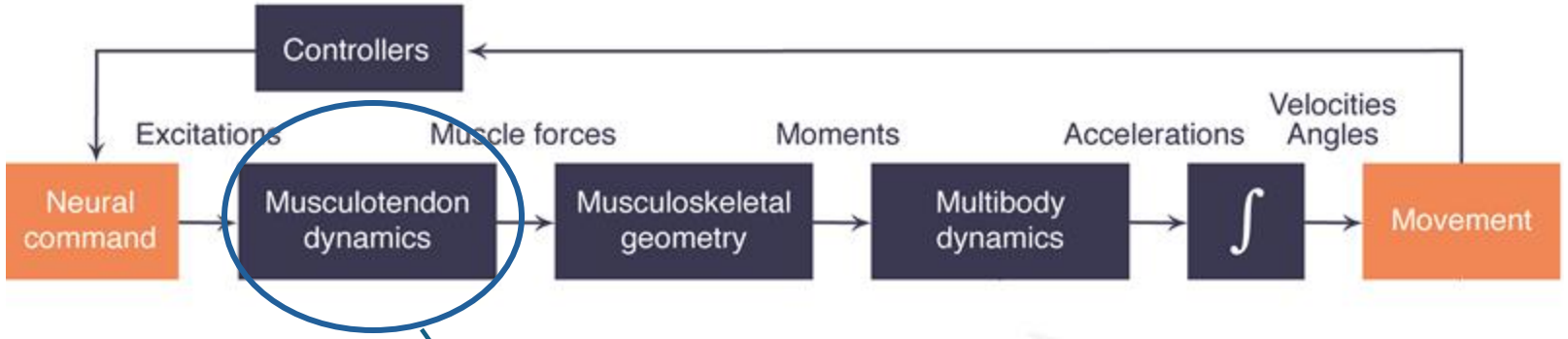
Recording EMG from the Skin Surface

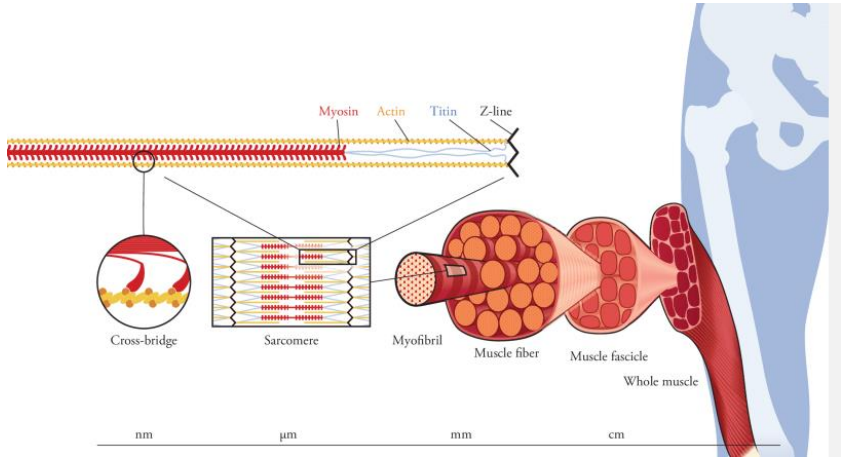
The EMG signal represents the sum of all the muscle fiber APs (measurable) from all of the active MUs that passes through the recording zone of the electrodes



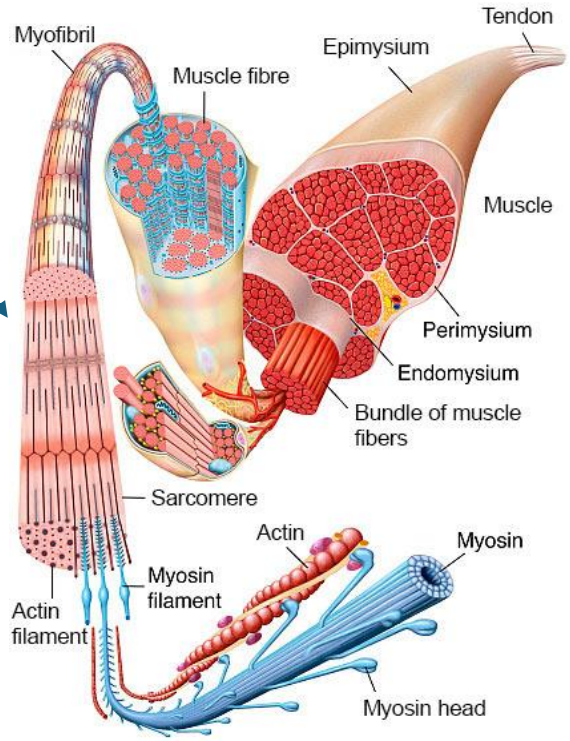
courtesy to Enrico Merlo, OT Bioelettronica for the illustration

From EMG to Muscle Forces





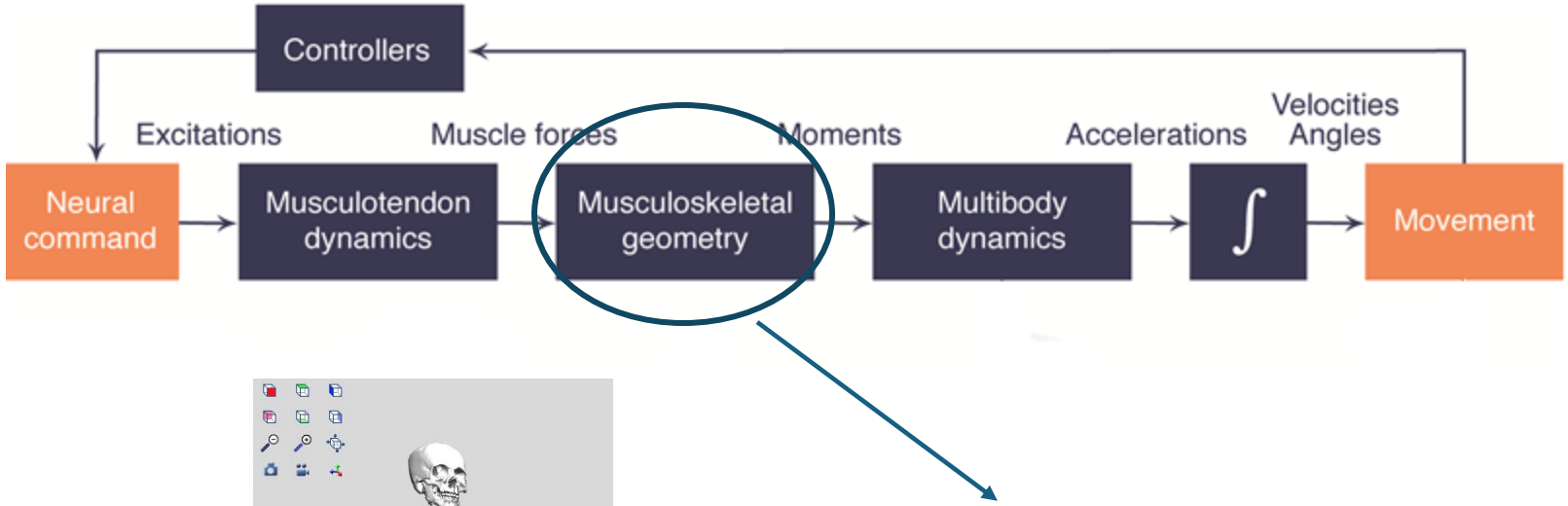
Multiscale structure of the skeletal muscle

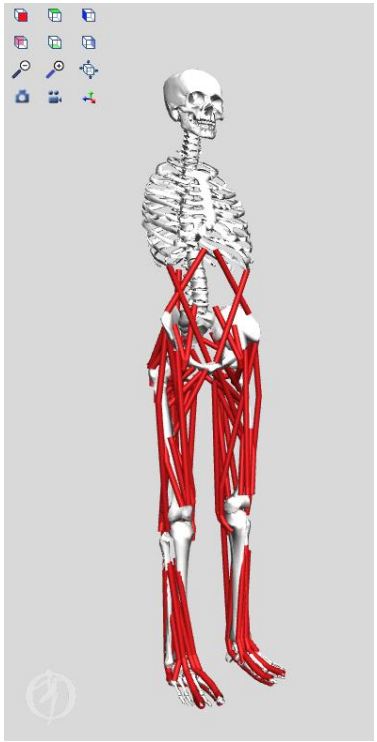




Muscle-tendon Unit (MTU)

From Forces to Moments

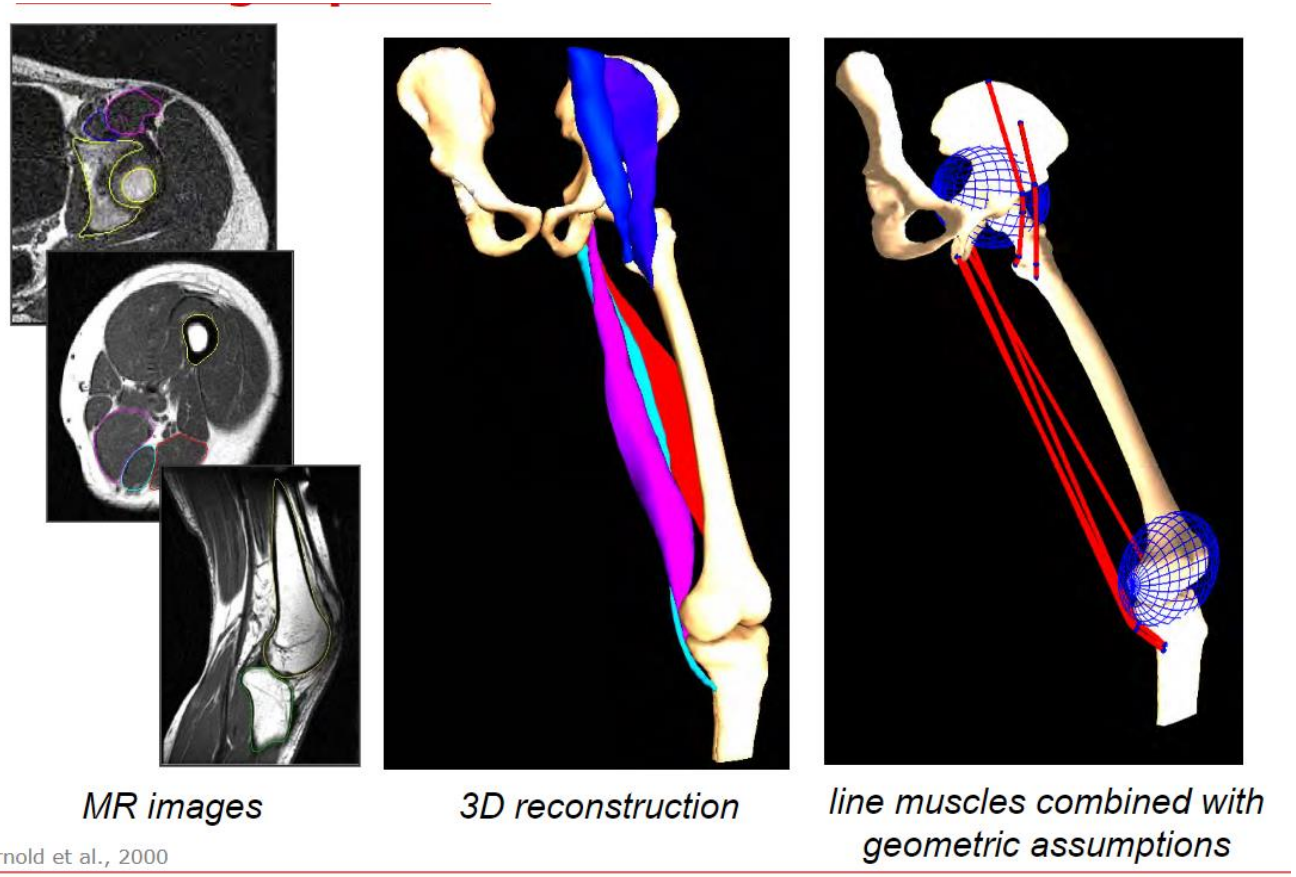




Transforms muscle forces to joint moments, major influence on muscle function

- A muscle's attachment points, path of action
- moment arms

Geometry: medical imaging



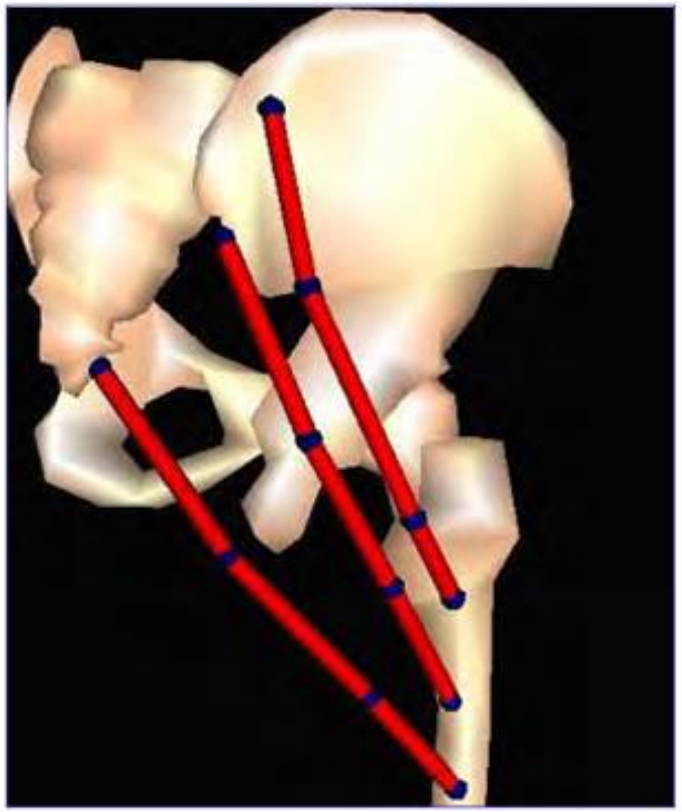
Courtesy to Jeff Reinbolt



Courtesy to Zhongzheng Wang

Geometry: medical imaging

Gluteus maximus and medius

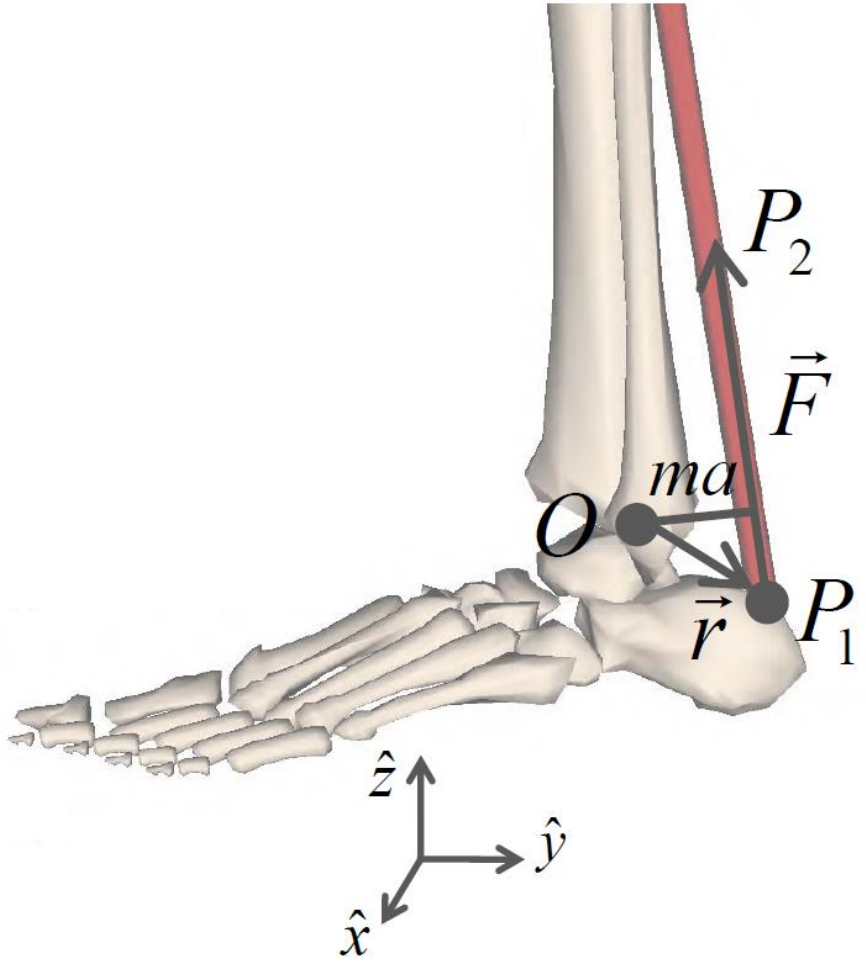




Moment Arm

The moment arm of $\vec{F}$ for $\hat{x}$ is defined as:

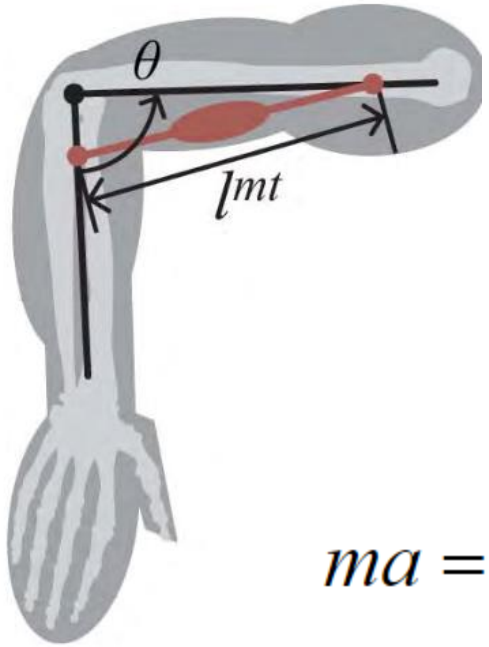
$$ma_x = \frac{\vec{r} \times \vec{F}}{|\vec{F}|} \bullet \hat{x}$$



Courtesy to Jeff Reinbolt

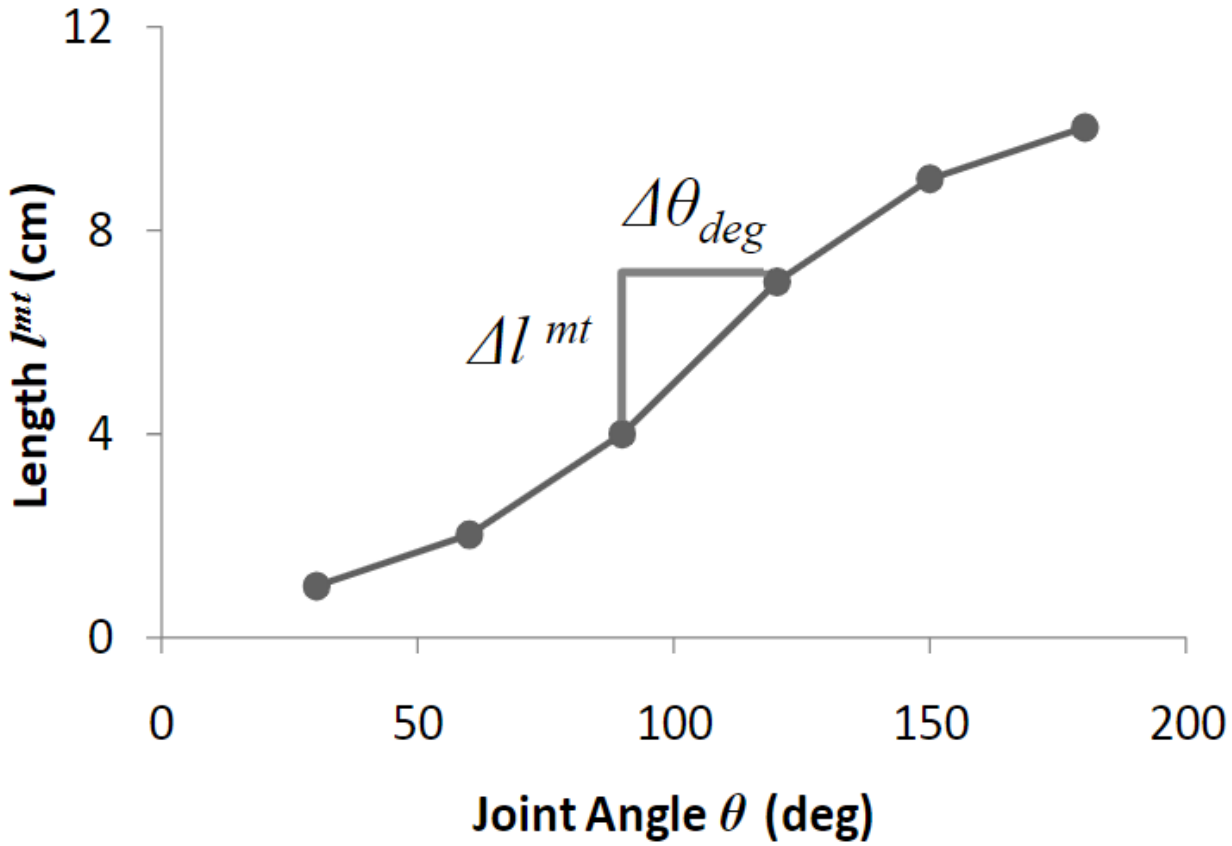
How do we measure moment arms?

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$$ma = \frac{\Delta l^{mt}}{\Delta \theta_{rad}}$$

The tendon excursion (TE) method

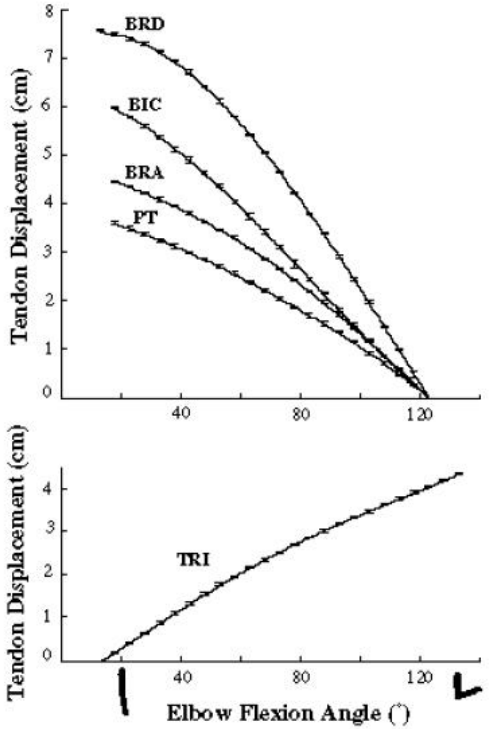


Courtesy to Jeff Reinbolt

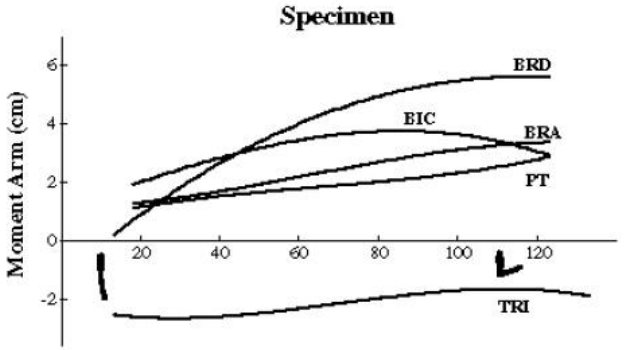
Example experimental results

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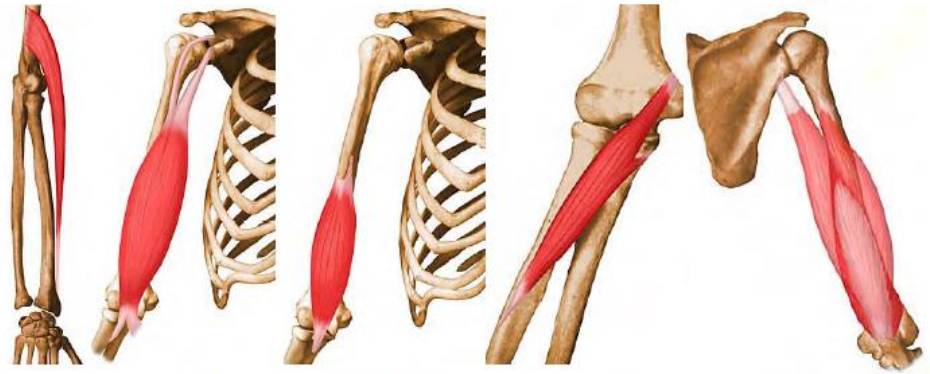
Displacements



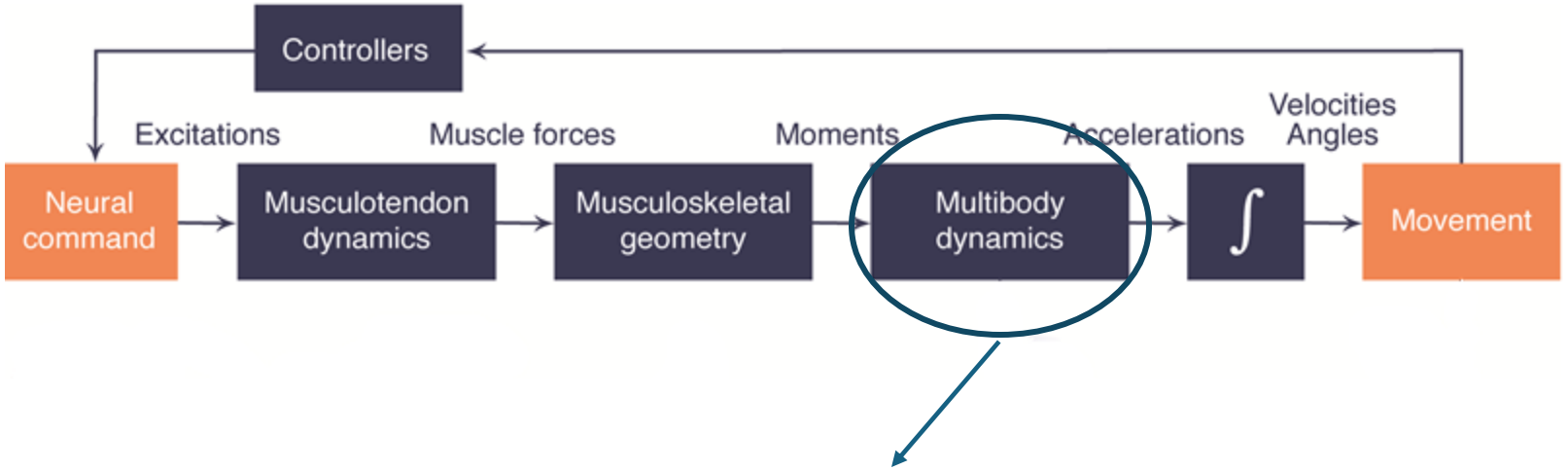
Moment Arms



University of Washington



Murray, Delp and Buchanan, 1994

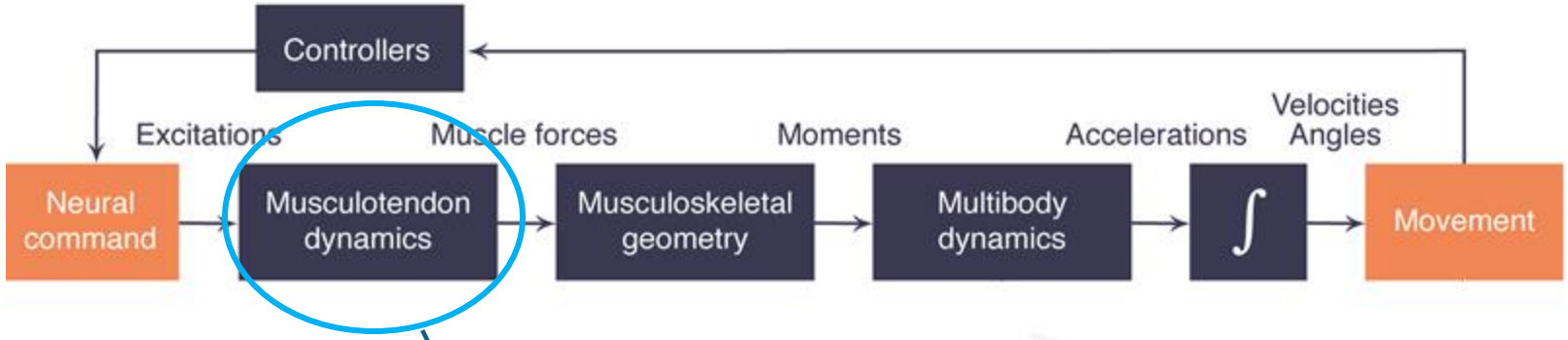


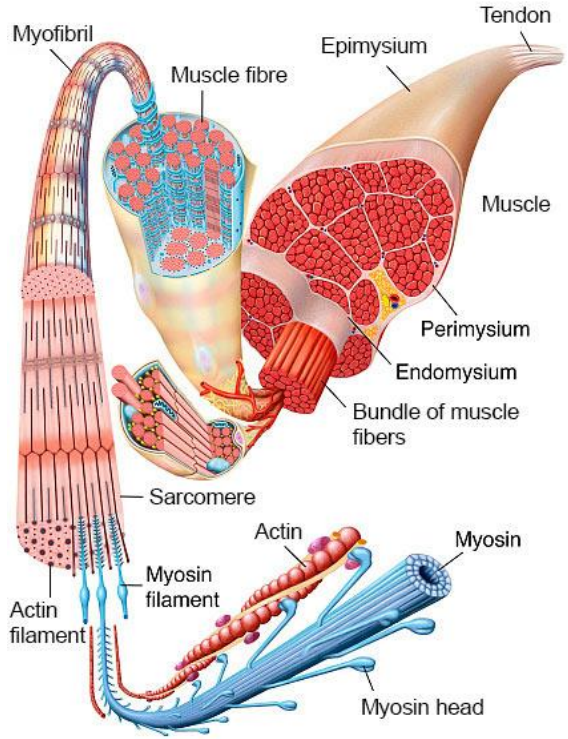
The equation of motion

$$\ddot{\vec{q}} = \vec{I}^{-1}(\vec{G} + \vec{C} + \vec{R} \cdot \vec{f}_{mt} + \vec{F}_{ext})$$

$\ddot{\vec{q}}$: vector of generalized accelerations
$\vec{I}$: matrix of segment masses and moments of inertia
$\vec{G}$: force vector of gravity
$\vec{C}$: vector of velocity related forces (Coriolis and centripetal)
$\vec{R}$: matrix of muscle moment arms
$\vec{f}_{mt}$: vector of muscle forces
$\vec{F}_{ext}$: vector of external forces

From EMG to Muscle Forces

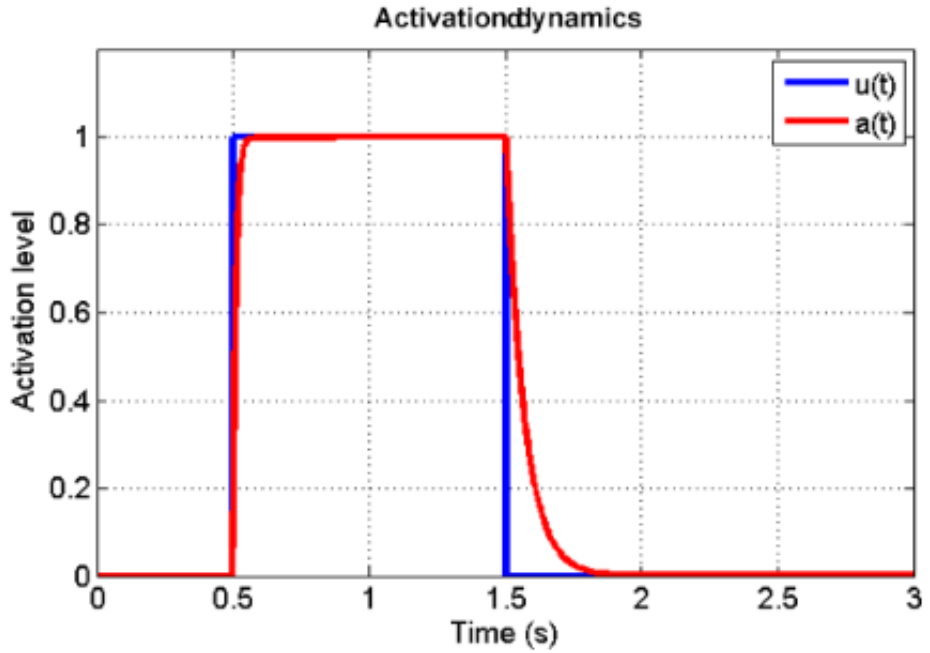




Musculotendon dynamics

Excitation-Activation Dynamics

Activation delay (5-10ms) < Deactivation delay (40-50ms)



excitation

$$\dot{a}(t) = (u(t) - a(t)) \cdot (t_1 u(t) + t_2)$$

activation

$1 \leq a \leq 0$

$1 \leq u \leq 0$

t_1 and t_2 are time constants

Francisco Romero-Sanchez et al., Frontiers in Neurorobotics, 2019

Musculotendon dynamics

Promobilia MoveAbility Lab

Muscle contraction dynamics

A simple and phenomenological mechanical model (Hill-type Model)

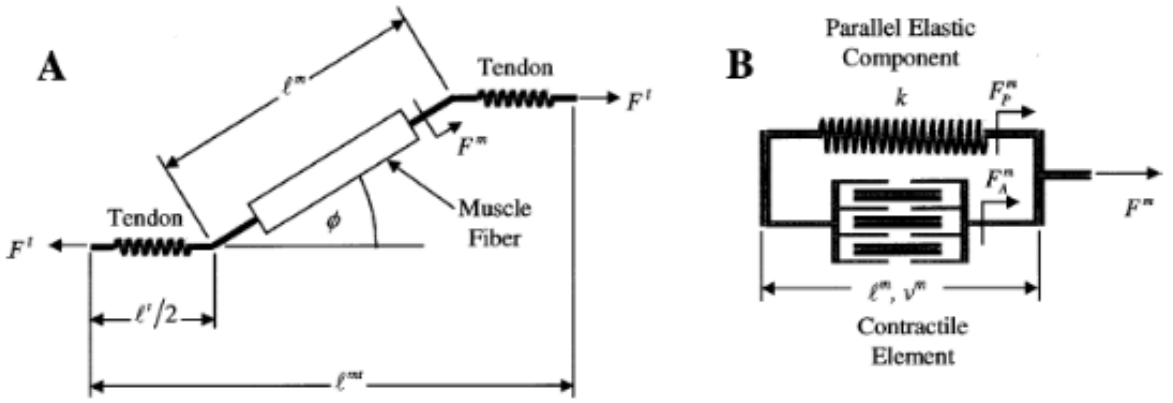


Figure 5.
(A) Schematic of muscle-tendon unit showing muscle fiber in series with the tendon. Note the pennation angle, φ, of the muscle fiber relative to the tendon and that the total tendon length, ℓt, is twice that of the tendon on either end of the muscle fiber, ℓt/2. (B) Schematic of muscle fiber with the contractile element and parallel elastic component. The nonlinear tendon stiffness is given by *k*. The force produced by the contractile element, *F*m, is a function of ℓm and vm, while the tendon force, *F*t, is a function of ℓt. The total muscle fiber force, *F*m, is the sum of *F*Am and *F*pm.

Thomas S. Buchanan et al., J Appl Biomech, 2004

Hill-type model

- Muscle fiber force

Maximal isometric muscle force

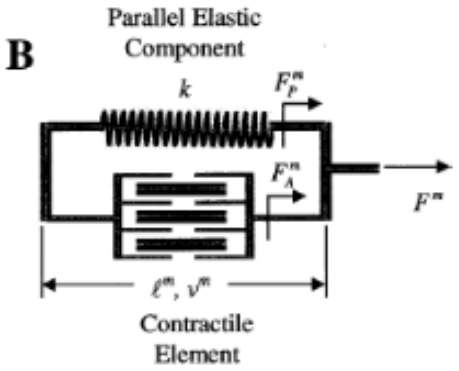
Active muscle force changes with length

Passive muscle force changes with length

$$F_M = F_M^0 \left[a f_{act}(\tilde{l}_M) f_v(\tilde{v}_M) + f_{pas}(\tilde{l}_M) \right]$$ Eq (1)

Muscle activation

Active muscle force changes with velocity

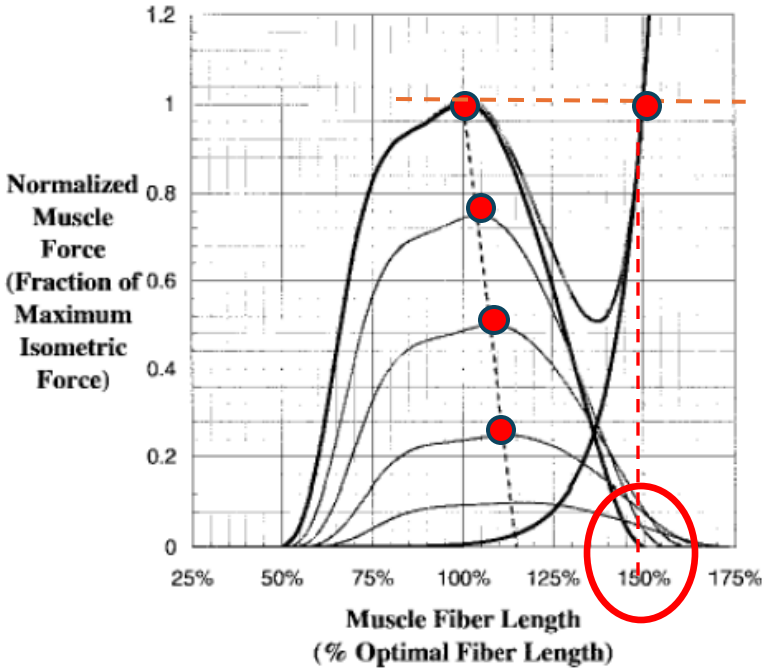


Thomas S. Buchanan et al., J Appl Biomech, 2004

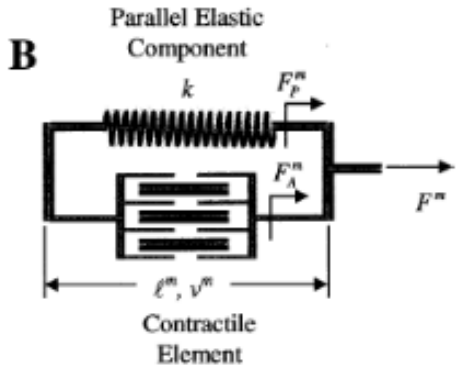
Hill-type model

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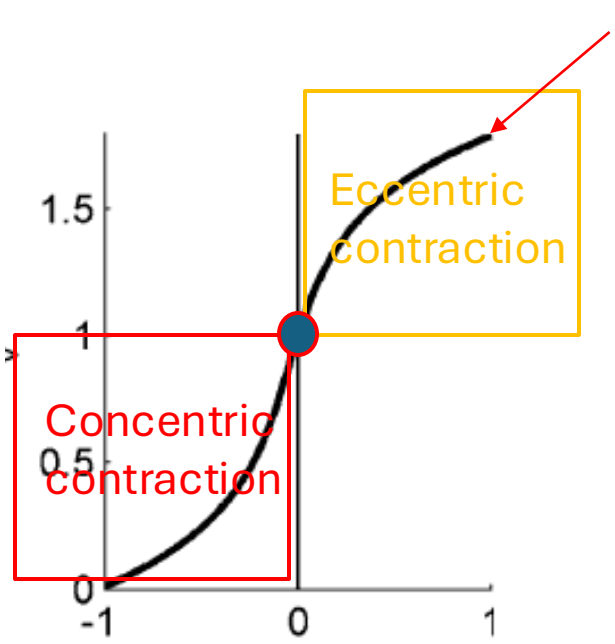
- Active and passive muscle force changes with length



$$\tilde{l}_{\mathrm{M}} = l_{\mathrm{M}} / l^{o}_{\mathrm{M}}$$



- Active muscle force changes with velocity



About 1.5 times of MIF

$$\tilde{v}_M$$

$$\tilde{v}_{\mathrm{M}} = v_{\mathrm{M}} / v_{\mathrm{M}}^{\max}$$

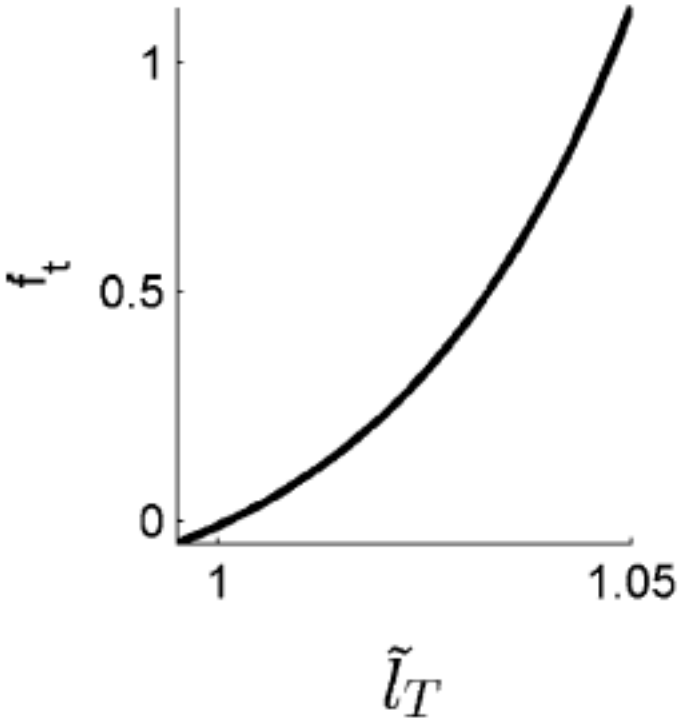
Tendon Dynamics

- Non-linear spring

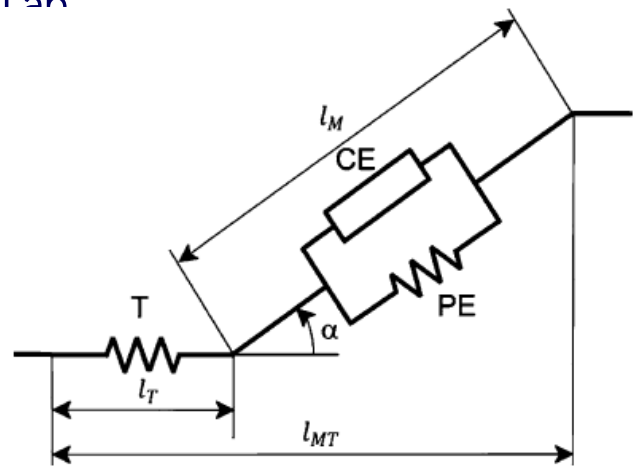
$$F_T = F_M^0 f_t(\tilde{l}_T)$$ Eq (2)

$$\tilde{l}_T = l_T / l_T^s$$

Tendon slack length



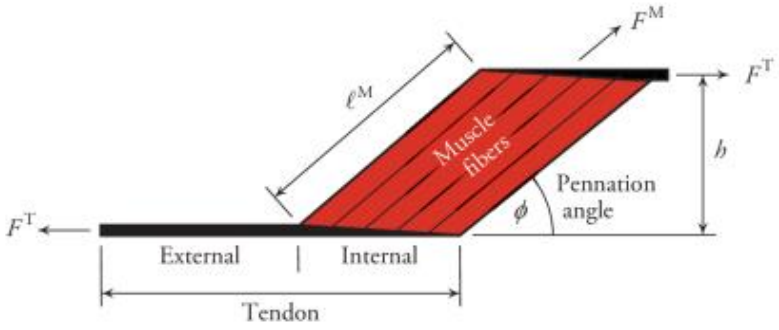
Muscle-tendon interaction



$$l_{MT} = l_T + l_M \cos\alpha, \qquad \text{Eq (3)}$$

$$l_M \sin\alpha = l_M^0 \sin\alpha_0, \qquad \text{Eq (4)}$$

$$F_T = F_M \cos\alpha. \qquad \text{Eq (5)}$$



Simplified geometric representation of a MTU

Feature parameters

- Optimal muscle fiber length
- Pennation angle at optimal fiber length
- Maximal isometric muscle force
- Maximal contraction velocity
- Tendon slack length

Group Exercise

Muscle with large moment arms undergo a larger change in length with joint rotation.

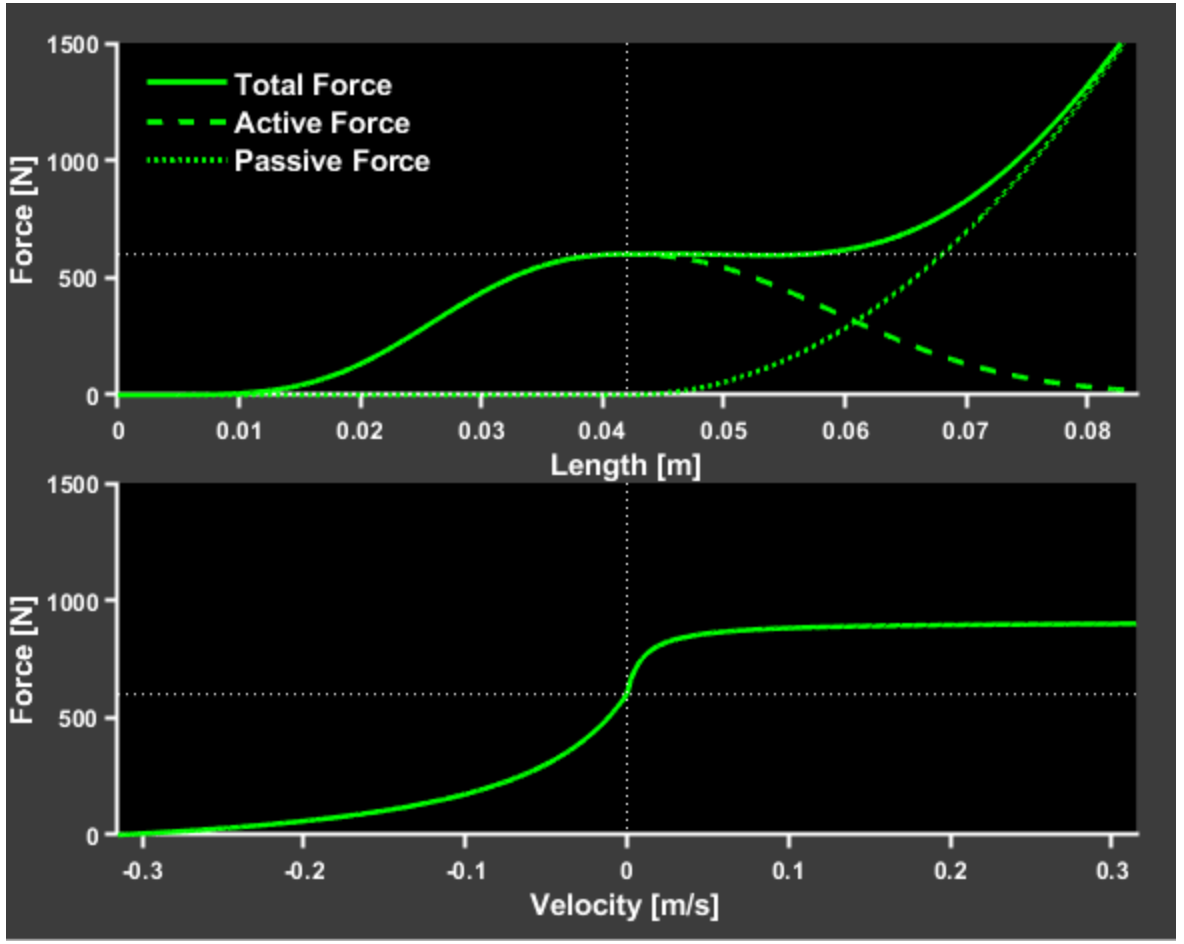
True or False?

Draw *and* ***Motivate*** *your answer*

Group Exercise

Promobilia MoveAbility Lab

Force-length and force-velocity curve for medial gastroc

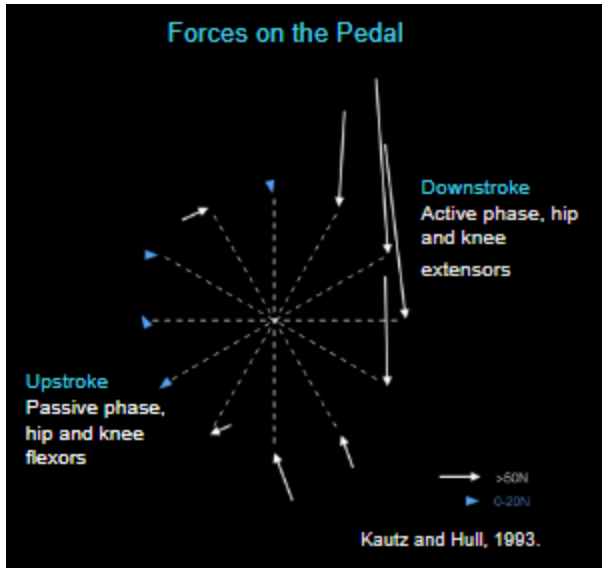


1. If the gastroc med's optimal fiber length is decreased, how will the new F-L and F-V curve look like?

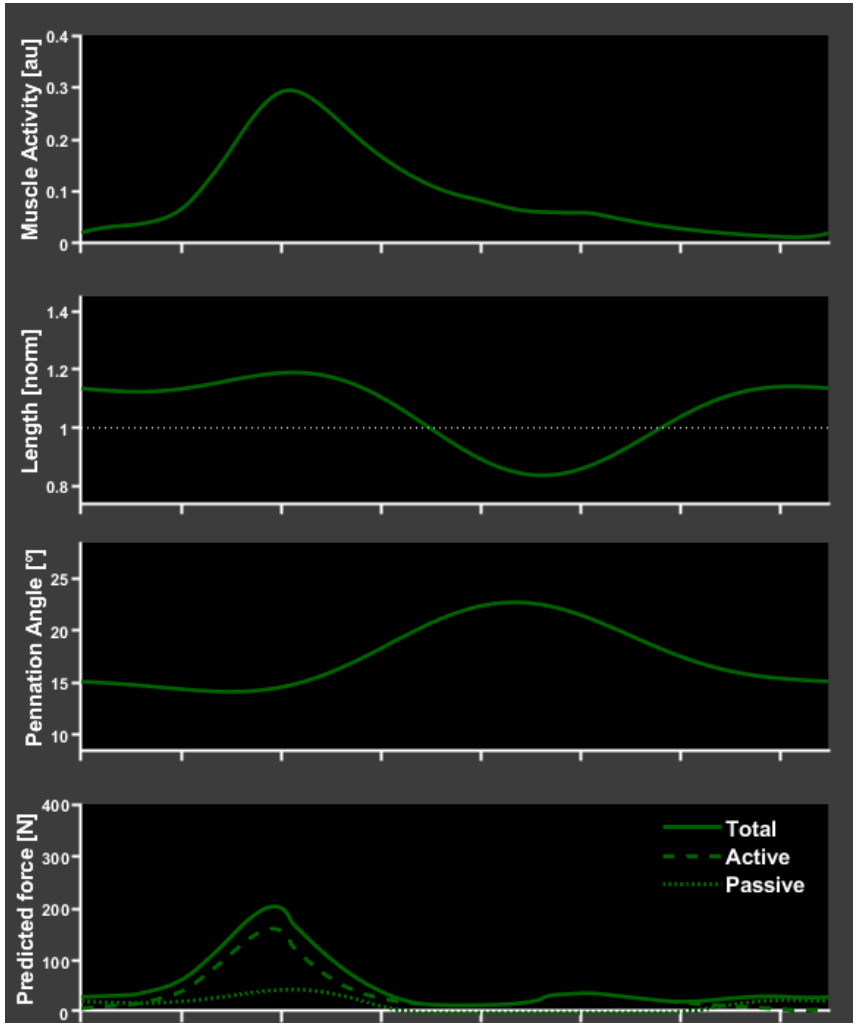
1. If the person has a weaker gastrocnemius, e.g. 20% weaker, how will the new F-L and F-V curve look like?

Draw *and* ***Motivate*** *your answer*

Group exercise



3. If the kinematics and activation are preserved in cycling, how will **normalized** fascicle length, the active and passive force curve look like in two different conditions?



Pedal angle (degrees)